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DEPI GRADUATION CAPSTONE PROJECT • SOFTWARE REQUIREMENTS SPECIFICATION

SOFTWARE REQUIREMENTS SPECIFICATION

Electricity Consumption Calculator

حاسبة استهلاك الكهرباء

Formal Baseline Engineering Requirements • Track: Mobile Application Development (Flutter)
Standards: IEEE Std 830-1998 • ISO/IEC/IEEE 29148:2018

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4. Introduction

4.1 Purpose of the Document

This Software Requirements Specification (SRS) establishes a formal, rigorous, and testable foundation for the engineering, quality assurance, and verification of the **Electricity Consumption Calculator** application. The document captures functional capabilities, non-functional quality attributes, mathematical models, data schemas, architectural trade-offs, and system interactions. It serves as the primary contractual and technical agreement between project stakeholders, university academic review panels, systems architects, mobile software engineers, and test specialists.

4.2 Document Conventions & Standards

- This specification adheres to the structural principles outlined in **IEEE Std 830-1998** (*Recommended Practice for Software Requirements Specifications*) and **ISO/IEC/IEEE 29148:2018** (*Systems and software engineering — Life cycle processes — Requirements engineering*).
- Requirement identification codes follow standard engineering taxonomy:
- `FR-XXX-NNN`: Functional Requirements, where `XXX` denotes the functional subsystem (e.g., `AUTH`, `METER`, `CALC`, `VOICE`, `OCR`, `CHART`, `BUDGET`, `APP`, `REC`, `TARIFF`, `PRIV`, `SET`) and `NNN` represents a zero-padded sequential counter.
- `NFR-XXX-NNN`: Non-Functional Requirements (e.g., `PERF`, `REL`, `SEC`, `USA`, `LOC`, `MNT`).
- `BR-NN`: Business Rules.
- `UC-NN`: Formal Use Cases.
- `TC-XXX-NNN`: Acceptance Test Specifications.
- Modal verbs conform to **IETF RFC 2119**:
- **SHALL / MUST**: Denotes an absolute, mandatory requirement for the initial baseline release.
- **SHOULD / RECOMMENDED**: Denotes a desirable feature or design preference that may be adjusted upon formal architectural review.
- **MAY / OPTIONAL**: Denotes a truly discretionary capability.

4.3 Intended Audience & Reading Suggestions

- **Academic Evaluators & Supervisors**: Focus on Section 4 (Scope), Section 9 (Non-Functional Requirements), Section 10 (Mathematical Models), Section 13 (System Diagrams), Section 16 (Traceability), and Section 18 (Architectural Trade-offs).
- **Mobile Software Development Engineers**: Read Section 7 (Functional Requirements), Section 8 (External Interfaces), Section 11 (Use Cases), Section 12 (Data Models), and Section 14 (Error Handling).
- **Quality Assurance & Verification Engineers**: Read Section 7 (Functional Requirements), Section 10 (Formulas & Numerical Examples), Section 14 (Edge Cases), Section 15 (Acceptance Criteria), and Section 16 (Traceability Matrix).
- **Product Owners & Business Analysts**: Review Section 5 (Product Overview), Section 6 (Scope Boundaries), Section 10 (Business Rules), and Section 18 (Open Questions).

4.4 Project Scope Overview

The **Electricity Consumption Calculator** is a specialized personal energy management mobile utility engineered for the Egyptian socio-economic context. It empowers individual household consumers to track electrical meter readings, convert reading differences into kilowatt-hour (kWh) consumption, estimate monthly bills using updateable progressive tariff tiers in Egyptian Pounds (EGP), record electrical appliances, establish spending budgets, and evaluate what-if scenarios to optimize consumption. The software incorporates voice-based numeric capture and optical character recognition (OCR) to minimize manual data entry barriers.

4.5 References & Regulatory Context

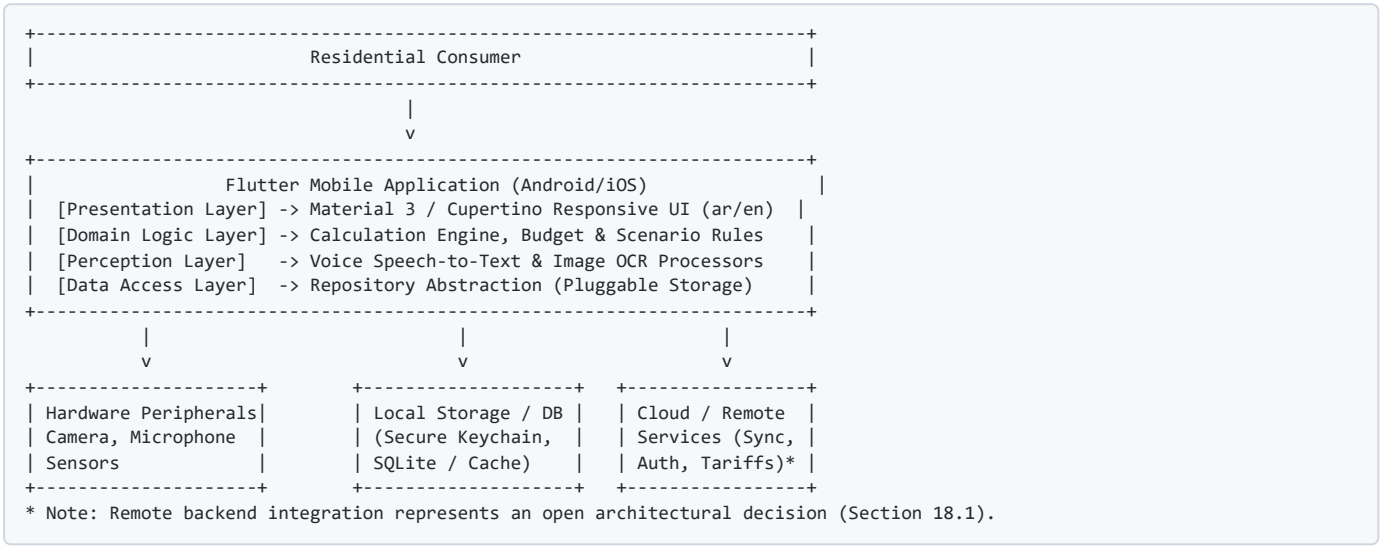
1. **IEEE Std 830-1998**: *IEEE Recommended Practice for Software Requirements Specifications*.
2. **ISO/IEC/IEEE 29148:2018**: *Systems and Software Engineering — Requirements Engineering*.
3. **EgyptERA Regulatory Framework**: *Egypt Electric Utility and Consumer Protection Regulatory Agency* (جهاز تنظيم مرفق الكهرباء وحماية المستهلك) guidelines on residential low-voltage retail tariff structures.

4. **W3C WCAG 2.1**: *Web Content Accessibility Guidelines* Level AA for mobile displays.
5. **Flutter Architecture Guide**: *Google Flutter Cross-Platform Development & State Management Standards*.

5. Overall Description

5.1 Product Perspective

The Electricity Consumption Calculator operates as an autonomous, user-centric mobile client running on the Flutter framework. Unlike utility-owned enterprise applications that integrate directly with smart grid telemetry or automated billing clearinghouses, this application functions as a **consumer-facing estimation, auditing, and planning tool**.



5.2 Product Functions Summary

1. **Account Lifecycle & Security**: Individual account registration, authentication, session management, and cascading account deletion with complete data isolation between users.
2. **Multi-Modal Reading Ingestion**: Support for manual keyboard entry, voice-based speech recognition (Egyptian Arabic and English), and optical character recognition (*text{OCR}*) from captured or uploaded meter photos.
3. **Consumption & Tiered Cost Engine**: Automated delta calculation (*text{kWh}*) between chronological readings, paired with a deterministic progressive tariff computation engine generating cost estimates in *text{EGP}*.
4. **Visual Analytics**: Interactive bar charts (consumption across days, months, or billing periods) and pie charts (appliance-level consumption breakdowns), accompanied by KPI cards and period-over-period comparisons.
5. **Budget Monitoring & Threshold Alerts**: Customizable periodic budget ceilings with real-time tracking, remaining balance calculations, run-rate end-of-period forecasting, and configurable early-warning triggers (*80*\% and *100*\%).
6. **Appliance Energy Modeling**: Inventory tracking of household appliances with rated wattage, voltage, current, power factor (*cos**phi*), operating schedules, and estimated individual/aggregate operating costs.
7. **Usage Optimization & Scenario Simulation**: Interactive scenario comparisons allowing users to simulate the financial and energy impacts of modifying appliance runtimes.
8. **Configurable Tariff Schedule**: Predefined, updateable tariff schedules supporting historical version retention to preserve estimation accuracy across regulatory rate adjustments.

5.3 User Classes & Characteristics

User Class	Description	Technical Literacy	Frequency of Use	Key Motivation
Residential Head of Household	Primary bill payer managing domestic utility expenses.	Moderate	2–4 times per month	Monitor monthly utility bills, avoid high progressive tariff brackets, and plan domestic budgets.
Budget-Constrained Consumer	Low-to-middle income household needing strict budget adherence.	Basic to Moderate	Daily to Weekly	Avoid crossing the *80*\% and *100*\% budget ceilings; identify heavy-drain appliances.

User Class	Description	Technical Literacy	Frequency of Use	Key Motivation
Energy Efficiency Enthusiast	Eco-conscious user optimizing appliance usage patterns.	High	Multiple times per week	Run scenario simulations, model standby power, and minimize domestic carbon footprint.
Non-Technical / Senior Citizen	User requiring simplified input mechanisms.	Novice	Monthly	Utilize voice input and camera OCR to avoid manual typing and small meter dials.

5.4 Operating Environment

- **Client Platforms**: Google Android (API Level 29 / Android 10.0 and higher) and Apple iOS (iOS 14.0 and higher).
- **Hardware Prerequisites**: Minimum 2 GB RAM, rear-facing camera with autofocus (minimum 8 MP recommended for OCR clarity), integrated microphone, active local storage (50 MB minimum application footprint).
- **Display Configurations**: Screen resolutions from $360 \times 640\text{ dp}$ to tablet dimensions, supporting both Portrait and Landscape orientations, with dynamic scaling for RTL (Arabic) and LTR (English).
- **Network Environment**: Fully operational in offline modes for manual entry, local calculations, and cached visualizations; network access required for cloud-based STT/OCR engines (if selected) or cloud synchronization.

5.5 Design & Implementation Constraints

- **Platform Technology**: The mobile client SHALL be constructed exclusively using the **Flutter** framework (Dart SDK) to maintain a unified cross-platform codebase.
- **Language & Cultural Context**: Full bidirectional interface support for Egyptian Arabic (`ar_EG`) with right-to-left (RTL) layout and English (`en_US`) with left-to-right (LTR) layout.
- **Data Sovereignty & Isolation**: Every record SHALL be strictly partitioned by authenticated user ID; multi-tenant data contamination is strictly prohibited.
- **Regulatory Independence**: The software SHALL NOT claim to issue legal, certified utility bills, nor represent itself as an official entity of the Egyptian Ministry of Electricity and Renewable Energy.
- **Non-Invention of Tariffs**: Tariff models SHALL rely on formal, configurable rule sets rather than hardcoded proprietary rate assumptions.

5.6 User Documentation & Onboarding

- The application SHALL incorporate an interactive first-use onboarding carousel explaining the core workflow (Reading $\rightarrow$ Consumption $\rightarrow$ Cost $\rightarrow$ Budget).
- Every major calculation screen SHALL provide accessible contextual tooltip descriptions explaining formulas, tier boundaries, and the distinction between actual meter usage and appliance approximations.
- In-app empty states SHALL guide new users to capture their baseline meter reading and register their primary appliances.

5.7 Assumptions & Dependencies

- Users possess physical or visual access to their domestic electricity meters (whether traditional electromechanical induction disc meters, electronic digital LCD meters, or prepaid card meters).
- Devices have appropriate operating system permissions granted for camera and microphone hardware.
- The underlying device operating system maintains an accurate system clock for timestamping readings.

6. Product Scope & Boundaries

6.1 In-Scope Capabilities (Initial Release)

1. Individual account registration, credential authentication, password recovery flows, and account deletion.
2. Ingestion of non-negative meter readings through manual entry, voice recognition, and image OCR.
3. Chronological reading validation, difference calculation (*text{kWh}*), and progressive tariff cost estimation (*text{EGP}*).
4. Appliance inventory recording, AC power-factor modeling, and schedule-based cost estimation.
5. Monthly and custom billing period budget tracking with *80*\% and *100*\% threshold warnings.
6. Visual analytics including time-series bar charts and appliance breakdown pie charts.
7. What-if scenario comparison tools for appliance runtime optimization.
8. Configurable tariff management supporting regulatory tier updates and historical tariff versioning.

6.2 Out-of-Scope Capabilities (Explicit Exclusions)

The following capabilities are explicitly excluded from the initial baseline release and SHALL NOT be assumed as present:

1. **Utility Backend Clearinghouse Integration**: No automated data synchronization with the Egyptian Electricity Holding Company (EEHC) or regional distribution companies (e.g., South Cairo Electricity Distribution, North Cairo, Alexandria).
2. **Smart Meter Hardware Telemetry**: No direct hardware coupling, Zigbee, optical probe reading, or smart meter IoT communication protocols.
3. **Electronic Bill Payment**: No embedded payment gateways (e.g., Fawry, Meeza, Vodafone Cash, credit card processing) for paying utility bills.
4. **Official Invoice Document Importation**: No parsing of scanned paper utility invoices or PDF bill statements.
5. **Multi-User Family Collaboration**: No shared household accounts, multi-tenant permission delegation, or multi-seat access in Version 1.0.0.
6. **Black-Box AI / Neural Load Disaggregation**: No non-intrusive load monitoring (NILM) based on high-frequency aggregate meter wave analysis.

6.3 System Boundary Context

The application is purely a **client-side personal decision-support system**. All calculations represent mathematical simulations and estimates designed to assist consumers in managing their electricity consumption.

7. Functional Requirements

7.1 FR-01: User Registration and Authentication

FR-AUTH-001: User Account Registration

- **Description**: The system SHALL allow a new user to register an account using a valid email address and a secure password.
- **Priority**: High (Must Have).
- **Inputs**: User full name, email address, password, password confirmation.
- **Processing Rules**:

1. Validate that the email address conforms to standard RFC 5322 format.
2. Validate that the password satisfies security complexity (minimum 8 characters, at least one uppercase letter, one lowercase letter, one numeric digit, and one special symbol).
3. Verify that the email address is not already registered in the system.
4. Generate a unique `user_id` (UUID v4) and hash the password using a salted cryptographic algorithm (Argon2id or PBKDF2 with SHA-256). Plaintext passwords SHALL NOT be stored under any circumstances.

- **Outputs**: Account creation confirmation message; initialization of default user settings; transition to authenticated session or verification screen.
- **Validation Rules**: Reject registration if required fields are missing, passwords do not match, or email format is malformed.
- **Acceptance Criteria**: Passwords meeting criteria succeed; duplicate emails fail with code `AUTH_ERR_DUPLICATE_EMAIL`.

FR-AUTH-002: User Authentication (Sign In)

- **Description**: The system SHALL authenticate an existing user using their registered credentials.
- **Priority**: High (Must Have).
- **Inputs**: Registered email address, password.
- **Processing Rules**:

1. Locate the account associated with the provided email.
2. Verify the provided password against the stored cryptographic hash.
3. Upon verification, generate a cryptographically signed session token containing the `user_id` and expiration timestamp.
4. Record the login event timestamp.

- **Outputs**: Authentication token; user profile state; redirection to the application dashboard.
- **Validation Rules**: If credentials do not match, increment failed login attempt counter. Display generic error: "*Invalid email or password*".
- **Acceptance Criteria**: Valid credentials navigate to the Dashboard within *1.5\text{ seconds}*; invalid credentials remain on login screen with clear error.

FR-AUTH-003: User Session Sign Out

- **Description**: The system SHALL allow an authenticated user to terminate their active session.
- **Priority**: High (Must Have).
- **Inputs**: Sign-out command triggered by user.
- **Processing Rules**: Invalidate the session token in secure local storage (and revoke on remote server, if cloud architecture is enabled). Clear in-memory user cache.
- **Outputs**: Navigation to the public Welcome/Login screen.
- **Acceptance Criteria**: Post-logout, navigating backward via hardware back button SHALL NOT display cached private user data.

FR-AUTH-004: Account Recovery (Password Reset)

- **Description**: The system SHALL provide an account recovery workflow for users who have forgotten their credentials.
- **Priority**: Medium (Should Have).

- **Inputs**: User registered email address.
- **Processing Rules**:

1. Verify if the email exists in the system.
2. If found, generate a cryptographically random, time-limited reset token (valid for 15 minutes).
3. Dispatch reset instructions (via email if cloud backend is present; or provide a local security-question recovery protocol if purely local).
4. Allow user to enter a new password meeting the complexity requirements of `FR-AUTH-001`.

- **Outputs**: Password reset confirmation; prompt to log in with new credentials.
- **Acceptance Criteria**: Expired tokens reject reset attempts; successfully updated passwords allow immediate sign-in.

FR-AUTH-005: Account and Data Deletion

- **Description**: The system SHALL allow an authenticated user to permanently delete their account and all associated records.
- **Priority**: High (Must Have - Privacy/Legal).
- **Inputs**: Explicit deletion request, current password re-authentication, explicit confirmation checkbox.
- **Processing Rules**:

1. Re-authenticate user credentials.
2. Execute cascading deletion across all relational tables: meter readings, meters, appliances, schedules, budgets, consumption estimates, and user settings tied to the `user_id`.
3. Purge cached local data, files, and stored session tokens.

- **Outputs**: Final deletion confirmation dialog; navigation to public landing screen.
- **Acceptance Criteria**: After deletion, querying the database for the given `user_id` yields zero records.

FR-AUTH-006: User Data Isolation & Ownership

- **Description**: The system SHALL strictly isolate each user's records such that no user can access, query, view, or alter another user's data.
- **Priority**: High (Must Have).
- **Processing Rules**: Every database query, whether local or remote, MUST append an explicit user ownership filter: `WHERE user_id = :authenticated_user_id`.
- **Acceptance Criteria**: Manual manipulation of local queries or API parameters with another user ID returns `403 Forbidden` or zero records.

7.2 FR-02: Electricity Meter Readings Management

FR-METER-001: Meter Entity Definition & Multi-Meter Support

- **Description**: The system SHALL allow users to register and manage one or more physical electricity meters (e.g., "Home Main Meter", "Workshop Meter", "Rental Apartment").
- **Priority**: Medium (Should Have).
- **Inputs**: Meter name, optional physical serial number, meter type (Mechanical Induction, Digital LCD, Prepaid Smart Card), initial baseline reading.
- **Processing Rules**: Associate meter entity with authenticated `user_id`. Assign unique `meter_id`.
- **Outputs**: Active meter profile.

FR-METER-002: Meter Reading Manual Ingestion

- **Description**: The system SHALL allow users to record a new meter reading manually via keyboard input.
- **Priority**: High (Must Have).
- **Inputs**: `meter_id`, numeric reading value (in *text{kWh}*), date and time of reading, optional notes.
- **Processing Rules**:

1. Validate that the value is a non-negative floating-point number.

2. Default timestamp to current system time if not specified.
3. Set `reading_source = 'manual'`.
4. Perform chronological consistency validation against existing readings for the same meter (`FR-METER-003`).

- **Outputs**: Confirmed reading record stored in persistent storage.

FR-METER-003: Chronological & Consistency Validation

- **Description**: The system SHALL validate that all meter readings maintain strict chronological and numeric consistency.
- **Priority**: High (Must Have).
- **Processing Rules**:

1. Locate the immediately preceding reading chronologically (*t_{prev} < t_{curr}*).
2. Assert that *Reading(t_{curr}) \ge Reading(t_{prev})*.
3. If *Reading(t_{curr}) < Reading(t_{prev})*, reject the reading with error `ERR_READING_BELOW_PREVIOUS` unless the user explicitly toggles a verified "Meter Replacement" or "Dial Rollover" flag (`FR-METER-006`).
4. Prevent duplicate reading entries with the exact same timestamp on the same meter.

- **Outputs**: Validation pass confirmation or descriptive validation error message.

FR-METER-004: Reading Modification and Correction

- **Description**: The system SHALL allow users to edit or delete existing meter readings to correct historical entry mistakes.
- **Priority**: Medium (Should Have).
- **Inputs**: Target `reading_id`, updated numeric value, updated timestamp, edit rationale notes.
- **Processing Rules**:

1. Re-validate chronological consistency with adjacent prior and subsequent readings.
2. Upon successful modification, trigger an automatic recalculation of consumption and cost estimates for affected billing periods (`FR-CALC-005`).

- **Outputs**: Updated reading record; updated historical consumption figures.

FR-METER-005: Reading Ingestion Metadata Tracking

- **Description**: The system SHALL record comprehensive audit metadata for every reading.
- **Priority**: High (Must Have).
- **Attributes Captured**: `reading_id`, `user_id`, `meter_id`, `reading_value`, `reading_timestamp`, `reading_source` (`manual`, `voice`, `ocr`), `notes`, `created_at`, `updated_at`.

FR-METER-006: Meter Rollover & Replacement Handling

- **Description**: The system SHALL support scenarios where an analog meter reaches its maximum mechanical dial capacity (e.g., *99,999\text{ kWh} \rightarrow 00,001\text{ kWh}*) or is physically replaced by the utility.
- **Priority**: Low (Could Have).
- **Processing Rules**:
- In dial rollover mode: *\Delta C = (\text{MaxDial} - M_{prev}) + M_{curr}*.
- In meter replacement mode: Record final reading on old meter, archive it, and establish initial reading on new meter without calculating negative consumption.

7.3 FR-03: Electricity Consumption and Cost Calculation

FR-CALC-001: Consumption Delta Calculation

- **Description**: The system SHALL calculate electricity consumption in kilowatt-hours (*\text{kWh}*) between two consecutive valid meter readings.
- **Priority**: High (Must Have).
- **Formula**:

$$\text{Consumption } (\text{kWh}) = M_{\text{current}} - M_{\text{previous}}$$

- **Outputs**: Precise consumption value rounded to two decimal places.

FR-CALC-002: Billing Period Assignment

- **Description**: The system SHALL associate calculated consumption with a distinct billing period defined by start date (t_{start}) and end date (t_{end}).
- **Priority**: High (Must Have).
- **Processing Rules**: If readings do not align perfectly with standard calendar month boundaries, the system SHALL calculate the exact elapsed duration in days and normalize daily average consumption:

$$\text{Daily Average Consumption} = \frac{\Delta C}{\text{Elapsed Days}}$$

FR-CALC-003: Progressive Tier Cost Calculation

- **Description**: The system SHALL calculate estimated electricity costs in Egyptian Pounds (EGP) by passing calculated consumption through the active, ordered progressive electricity tariff tiers (`FR-10`).
- **Priority**: High (Must Have).
- **Processing Rules**:

1. Query active tariff configuration corresponding to the reading period.
2. Sequentially allocate consumption across defined tier brackets $[T_{\text{start}}, T_{\text{end}})$.
3. Multiply allocated bracket kWh by the corresponding tier rate (EGP/kWh).
4. Sum the cost across all applicable brackets.

- **Outputs**: Total estimated cost (EGP), itemized tier breakdown showing kWh and subtotal per tier.

FR-CALC-004: Estimation Disclaimer Labeling

- **Description**: The system SHALL visibly and unambiguously label all calculated monetary amounts as **"Estimated Cost / تكلفة تقديرية"**.
- **Priority**: High (Must Have).
- **Processing Rules**: Accompany all cost displays with a standard user-facing disclaimer: "*This cost is an estimate based on recorded meter readings and configured tariff rates. It does not replace or constitute an official utility invoice issued by the distribution company.*"

FR-CALC-005: Dynamic Recalculation on Data Mutation

- **Description**: When a meter reading is added, modified, or deleted, or when the underlying tariff configuration is updated, the system SHALL recalculate affected historical consumption estimates.
- **Priority**: High (Must Have).

7.4 FR-04: Voice-Based Numeric Input

FR-VOICE-001: Audio Capture & Permission Management

- **Description**: The system SHALL provide a dedicated microphone control button on meter reading and numeric input forms.
- **Priority**: Medium (Should Have).
- **Processing Rules**:

1. Upon interaction, inspect OS microphone runtime permission.
2. If permission is ungranted, display an informative permission rationale dialog explaining the utility of voice capture.
3. If permission is denied, gracefully fall back to manual keyboard input without impeding user progress.

- **Acceptance Criteria**: Denying permission does not crash the app; keyboard input is immediately focused.

FR-VOICE-002: Bilingual Speech-to-Text Transcription

- **Description**: The system SHALL transcribe spoken audio into text supporting both Egyptian Arabic (`ar_EG`) and English (`en_US`).
- **Priority**: Medium (Should Have).
- **Processing Rules**: Transcribe spoken numbers, handling both digit sequences (e.g., "واحد اثنين اربعة خمسة" / "one two four five") and cardinal values (e.g., "اثنا عشر ألف وأربعمائة وخمسون" / "twelve thousand four hundred fifty").

FR-VOICE-003: Numeric Token Parsing & Decimal Handling

- **Description**: The system SHALL extract a clean, valid floating-point number from transcribed speech.
- **Priority**: Medium (Should Have).
- **Processing Rules**:

1. Parse colloquial decimal indicators (e.g., "نقطة" ,"فاصلة", "point", "comma").
2. Normalize Eastern Arabic numerals (٩-•) to standard decimal digits (0-9).
3. Discard extraneous conversational filler words (e.g., "العداد النهاردة", "the reading is").

FR-VOICE-004: User Review & Confirmation Gate

- **Description**: Voice recognition output SHALL NEVER be committed directly to persistent storage as an authoritative reading without explicit user confirmation.
- **Priority**: High (Must Have).
- **Processing Rules**: Populate the extracted number into an editable text field within a confirmation modal, allowing the user to review, manually edit, re-record, or confirm the reading.

7.5 FR-05: OCR-Based Meter Reading Extraction

FR-METER-OCR-001: Image Acquisition (Camera & Gallery)

- **Description**: The system SHALL enable users to capture a live photograph of their electricity meter using the device camera or select an existing photo from the local image gallery.
- **Priority**: Medium (Should Have).
- **Processing Rules**: Verify runtime camera/storage permissions; launch native camera viewfinder with an overlay framing guide outlining the meter's numeric register.

FR-METER-OCR-002: Optical Character Recognition Execution

- **Description**: The system SHALL process the acquired meter image through an optical character recognition (*text{OCR}*) engine to extract numeric text.
- **Priority**: Medium (Should Have).
- **Processing Rules**: Apply image preprocessing (orientation correction, grayscale conversion, and contrast enhancement); execute text detection; extract bounding boxes containing numeric strings.

FR-METER-OCR-003: Candidate Filtering & Disambiguation

- **Description**: The system SHALL filter detected text blocks to isolate plausible meter readings from peripheral numbers (e.g., meter serial numbers, manufacturer model codes, voltage ratings like *220\text{V}*).
- **Priority**: Medium (Should Have).
- **Processing Rules**:

1. Match candidate strings against valid meter display patterns (typically 4 to 7 digits, optionally followed by 1 decimal digit).
2. If multiple candidate numeric blocks are detected with similar confidence, present the candidates visually with image crop thumbnails and require the user to tap the intended reading.
3. Do not blindly select an arbitrary number when ambiguity exists.

FR-METER-OCR-004: Extraction Review, Manual Edit, and Source Tagging

- **Description**: The extracted OCR value SHALL be presented in an editable confirmation interface alongside the captured photo thumbnail.
- **Priority**: High (Must Have).
- **Processing Rules**: Require explicit user confirmation; upon saving, record `reading_source = 'ocr'` in the reading entity metadata.

7.6 FR-06: Consumption Visualization & Analytics

FR-CHART-001: Historical Consumption Bar Charts

- **Description**: The system SHALL render interactive bar charts illustrating electricity consumption (kWh) grouped across selectable time dimensions: Daily, Monthly, or Billing Period.
- **Priority**: High (Must Have).
- **Data Source**: Authenticated user's validated historical meter readings and calculated consumption estimates.
- **UI Attributes**: Visual axis labels, value callouts on bar tap, unit indicators (kWh), and chronological sorting.

FR-CHART-002: Appliance Distribution Pie / Donut Chart

- **Description**: The system SHALL render a pie or donut chart displaying the proportional distribution of estimated electricity consumption among registered household appliances.
- **Priority**: Medium (Should Have).
- **Data Source**: Aggregated monthly consumption calculated from registered appliance profiles (`FR-08`).
- **Processing Rules**: Group appliances contributing $< 3\%$ of total consumption into a consolidated "Other / أخرى" slice to preserve visual legibility.

FR-CHART-003: Period-over-Period Comparative Metrics

- **Description**: The system SHALL compute and display comparative indicators contrasting the current billing period against the immediately preceding period.
- **Priority**: Medium (Should Have).
- **Indicators**: Absolute difference (ΔkWh, ΔEGP) and percentage change ($\pm\%$). Render directional trend arrows (green for reduction, red for increase).

FR-CHART-004: Empty States & Insufficient Data Handling

- **Description**: When a user possesses zero or insufficient historical records (e.g., fewer than two readings for bar charts, or no registered appliances for pie charts), the system SHALL render an educational empty-state illustration with an actionable prompt (e.g., "*Add your second meter reading to unlock consumption charts*").
- **Priority**: High (Must Have).

7.7 FR-07: Electricity Budget Management

FR-BUDGET-001: Budget Creation and Configuration

- **Description**: The system SHALL allow users to establish an electricity spending ceiling in Egyptian Pounds (EGP) for a defined billing cycle.
- **Priority**: High (Must Have).
- **Inputs**: Budget amount (EGP), billing cycle start date (day of month, e.g., 1st or 15th), custom duration in days.
- **Validation**: Budget amount MUST be a positive number > 0.

FR-BUDGET-002: Real-Time Spending Tracking

- **Description**: The system SHALL calculate and display real-time budget utilization metrics:
- Total Budget Allocated (EGP)

- Estimated Spending to Date (*text{EGP}*)
- Remaining Budget Balance (*text{EGP}*)
- Percentage of Budget Consumed (\%)
- **Priority**: High (Must Have).

FR-BUDGET-003: Configurable Early-Warning Threshold Alerts

- **Description**: The system SHALL evaluate budget utilization against configurable warning thresholds (defaulted to *80*\% warning and *100*\% limit).
- **Priority**: High (Must Have).
- **Processing Rules**:

1. When spending reaches or exceeds *80*\% of budget, display a prominent yellow warning banner on the dashboard.
2. When spending reaches or exceeds *100*\%, display an alert indicating budget overrun.
3. Threshold alerts SHALL NOT claim to be official notifications from the electric utility company.

FR-BUDGET-004: End-of-Period Run-Rate Spending Projection

- **Description**: The system SHALL compute a projected month-end expenditure based on the current consumption run-rate:

$$\text{Projected Cost} = \left(\frac{\text{Cost to Date}}{\text{Elapsed Days in Cycle}}\right) \times \text{Total Days in Cycle}$$

- **Priority**: Medium (Should Have).
- **Processing Rules**: Require at least 5 elapsed days within the billing period before displaying projection to prevent extreme statistical variance.

7.8 FR-08: Electrical Appliance Management

FR-APP-001: Appliance Inventory Recording

- **Description**: The system SHALL allow users to maintain an inventory of domestic electrical appliances.
- **Priority**: High (Must Have).
- **Attributes**: `appliance_id`, `user_id`, `name` (e.g., "Living Room AC"), `category` (Cooling, Heating, Refrigeration, Lighting, Entertainment, Cooking, Other), `rated_power_watts` (*W*), `rated_voltage_volts` (*V*), `rated_current_amperes` (*A*), `power_factor` (*cos\phi*), `quantity`, `standby_power_watts`, `notes`.

FR-APP-002: Operating Schedule Definition

- **Description**: For each appliance, the system SHALL allow users to record typical operating schedules:
- Average daily operating hours (range: *0.0* to *24.0**text{ hours/day}*)
- Operating days per billing cycle (range: *1* to *31**text{ days/month}*)
- **Priority**: High (Must Have).
- **Validation Rules**: Daily hours MUST NOT exceed *24.0**text{ hours}*. Total monthly hours MUST NOT exceed *text{Days in Month} \times 24.0*.

FR-APP-003: Individual & Aggregate Energy Computation

- **Description**: The system SHALL compute estimated periodic energy consumption for each appliance:

$$E_{\text{single}} (\text{kWh}) = \frac{P_{\text{watts}} \times \text{Hours/Day} \times \text{Days/Month}}{1000}$$

$$E_{\text{total}} (\text{kWh}) = E_{\text{single}} \times \text{Quantity}$$

- **Priority**: High (Must Have).

FR-APP-004: Electrical Specification Resolution Rules

- **Description**: If the user inputs rated power in Watts (*W*), the system SHALL utilize that value directly. If the user provides voltage (*V*) and current (*A*) but omits wattage, the system SHALL evaluate power using active AC power equations (`BR-06`), requiring or defaulting an appropriate power factor (*cos*phi*).
- **Priority**: High (Must Have).

FR-APP-005: Standby / Vampire Power Tracking

- **Description**: The system SHALL support an optional standby power rating (*W*) applied during non-operating hours:

$$E_{\text{standby}} (\text{kWh}) = \frac{P_{\text{standby}} \times (24 - \text{Operating Hours}) \times \text{Days}}{1000}$$

- **Priority**: Low (Could Have).

7.9 FR-09: Usage Optimization Recommendations

FR-REC-001: Deterministic Optimization Engine

- **Description**: The system SHALL analyze registered appliances, their operating hours, estimated consumption, and the active budget to generate actionable reduction recommendations.
- **Priority**: Medium (Should Have).
- **Processing Rules**: The engine SHALL utilize deterministic, transparent heuristic rules rather than unverified black-box machine learning algorithms.

FR-REC-002: High-Drain Appliance Identification

- **Description**: The system SHALL automatically rank and flag appliances contributing > *20*\% of the aggregate estimated consumption (e.g., air conditioners, electric water heaters, space heaters).
- **Priority**: Medium (Should Have).

FR-REC-003: What-If Scenario Simulation

- **Description**: The system SHALL provide an interactive simulation interface allowing users to adjust appliance runtimes (e.g., "*Reduce Living Room AC from 8 hours/day to 6 hours/day*") and observe the simulated savings in both *\text{kWh}* and *\text{EGP}*.
- **Priority**: Medium (Should Have).
- **Outputs**: Delta *\Delta \text{kWh}* saved, Delta *\Delta \text{EGP}* saved, updated tier allocation.
- **Disclaimer**: Label savings explicitly as "*Simulated Potential Savings / وفورات تقديرية محتملة*".

7.10 FR-10: Tariff Management & Update Mechanism

FR-TARIFF-001: Structured Tariff Schedule Representation

- **Description**: The system SHALL maintain a structured tariff configuration entity containing:
- `tariff_id` , `version_name` , `effective_from_date` , `effective_to_date` , `customer_category` (Residential, Commercial), `is_active` .
- Ordered child tiers: `tier_order` , `tier_start_kwh` , `tier_end_kwh` , `rate_per_kwh_egp` , `tier_label` .
- **Priority**: High (Must Have).

FR-TARIFF-002: Historical Tariff Version Preservation

- **Description**: The system SHALL preserve all historical tariff schedules. Modifying or introducing a new tariff schedule SHALL NOT overwrite previous versions.
- **Priority**: High (Must Have).

- **Processing Rules**: When calculating or reviewing past billing periods, the system SHALL retrieve the tariff version whose effective date range encompasses that period.

FR-TARIFF-003: Controlled Tariff Update Pipeline

- **Description**: The system SHALL support updating tariff schedules without requiring hardcoded application binary recompilation (e.g., via administrative JSON remote config delivery or controlled in-app tariff profile import).
- **Priority**: High (Must Have).

FR-TARIFF-004: Non-Invention of Official Rates

- **Description**: The software requirements and system data SHALL NOT invent or fabricate official Egyptian government utility rates. Any sample tariffs used during testing or default onboarding SHALL be explicitly watermarked and labeled as **"Sample Illustrative Tariff — Requires Verification against Official EgyptERA Schedules"**.
- **Priority**: High (Must Have).

7.11 FR-11: User Data Isolation and Privacy

FR-PRIV-001: Row-Level Tenant Partitioning

- **Description**: All user data entities (meters, readings, appliances, budgets, estimates) MUST possess a mandatory, immutable foreign key reference to `user_id`.
- **Priority**: High (Must Have).

FR-PRIV-002: Local Storage Cryptographic Protection

- **Description**: When stored locally on the client device, sensitive authentication credentials, tokens, and private records SHALL be encrypted at rest using platform-standard cryptographic keychains (Android EncryptedSharedPreferences / KeyStore, iOS Keychain).
- **Priority**: High (Must Have).

FR-PRIV-003: Ephemeral Processing of Perception Media

- **Description**: Captured meter images and voice audio recordings utilized for OCR and speech recognition SHALL be processed ephemerally. Raw image bitmaps and audio clips SHALL NOT be transmitted to third parties or retained permanently unless explicitly authorized by the user for diagnostic bug reporting.
- **Priority**: High (Must Have).

7.12 FR-12: Application Settings and History Management

FR-SET-001: Language & Locale Preferences

- **Description**: The system SHALL allow users to switch application language between Egyptian Arabic (`ar_EG`) and English (`en_US`). The UI SHALL dynamically re-render layout orientation (*text{RTL}* *leftrightarrow* *text{LTR}*) immediately upon selection without requiring application restart.
- **Priority**: High (Must Have).

FR-SET-002: Currency & Formatting Customization

- **Description**: The system SHALL format all currency values in Egyptian Pounds (*text{EGP}* / ج.م) adhering to Egyptian locale standards, supporting user configuration for digit presentation (Western Arabic `0-9` vs Eastern Arabic `٩-٠`).
- **Priority**: Medium (Should Have).

FR-SET-003: Comprehensive Reading History Log

- **Description**: The system SHALL provide a paginated, filterable audit log of all recorded meter readings, displaying date, value, delta consumption, origin source (manual/voice/OCR), and associated notes.
- **Priority**: High (Must Have).

8. External Interface Requirements

8.1 User Interfaces (Flutter Client)

- **Design Paradigm**: Implementation using Flutter's Material 3 (Material You) and Cupertino widget architectures, providing modern, tactile mobile interactions.
- **Responsive Layout**: Fluid layout grids accommodating screen densities from low-density mobile displays (320 dpi) to high-density flagship devices and tablets.
- **Bidirectionality**: Complete support for Right-to-Left (RTL) layout mirroring for Arabic and Left-to-Right (LTR) for English, including directional icon flipping and text alignment.
- **Visual Feedback**: Distinct micro-interactions for button presses, visual audio waveforms during voice capture, bounding-box overlays during OCR capture, and skeleton loaders during chart rendering.

8.2 Hardware Interfaces

- **Camera Sensor Interface**: Access to device camera hardware via the Flutter `camera` or `image_picker` platform channels. Requires autofocus, flashlight/torch toggle control (to illuminate dark meter utility closets), and minimal frame capture latency.
- **Microphone Hardware Interface**: Audio input stream capture at a minimum sampling rate of 16 kHz mono (16-bit PCM) for speech-to-text processing.
- **Local Persistent Storage**: Flash memory storage interface for local database files, shared preferences, and cached assets.

8.3 Software Interfaces

- **Speech Recognition Engine**:
- *Candidate Option A (On-Device)*: Native OS Speech Recognizer (Android `SpeechRecognizer` / iOS `SFSpeechRecognizer`). Trade-off: High privacy, zero bandwidth, but variable dialect support for Egyptian colloquial Arabic.
- *Candidate Option B (Cloud Service)*: Remote STT API (e.g., Google Cloud Speech-to-Text / Whisper). Trade-off: Superior Egyptian Arabic accuracy, but requires internet connection and incurs per-call latency/costs.
- **Optical Character Recognition Engine**:
- *Candidate Option A (On-Device)*: Google ML Kit Text Recognition v2. Trade-off: Completely offline, fast ($< 500\text{ms}$), free, but limited accuracy on highly reflective curved analog glass dials.
- *Candidate Option B (Cloud OCR)*: Cloud Vision API. Trade-off: High accuracy on distorted analog dials, but requires network connectivity and external API credentials.
- **Data Repository Interface**: A strict Dart abstract repository contract (`IUserRepository` , `IMeterRepository` , `ITariffRepository`) decoupling presentation blocs from the concrete persistence driver (SQLite, Hive, Supabase, or REST).
- **Data Visualization Library**: Vector chart rendering engine (e.g., `fl_chart`) capable of rendering responsive bar, pie, and line graphs natively on the Skia/Impeller graphics pipeline.

8.4 Communications Interfaces

- **Network Protocol**: HTTPS over TLS 1.3 for all outbound remote API communications (if cloud or remote config services are active).
- **Payload Format**: Application JSON (`Content-Type: application/json`) with strict UTF-8 character encoding.
- **Security Headers**: Standard bearer token authorization (`Authorization: Bearer`) with payload integrity checks.

9. Non-Functional Requirements

9.1 Performance Requirements

- **NFR-PERF-001 (Cold Startup)**: The mobile application SHALL achieve cold startup to an interactive dashboard in $< 2.0\text{ seconds}$ on mid-tier hardware (e.g., Octa-core 2.0 GHz, 4 GB RAM).
- **NFR-PERF-002 (Calculation Latency)**: The core consumption and progressive tariff cost calculation engine SHALL complete execution in $< 50\text{ milliseconds}$ for any given billing period.
- **NFR-PERF-003 (Visualization Rendering)**: Interactive bar and pie charts SHALL render completely in $< 300\text{ milliseconds}$ upon screen navigation.
- **NFR-PERF-004 (On-Device OCR Latency)**: When utilizing on-device OCR, text detection from a captured bitmap SHALL yield extracted candidates in $< 1.5\text{ seconds}$.
- **NFR-PERF-005 (Memory Footprint)**: Client runtime memory utilization SHALL NOT exceed 180 MB under peak graphical and camera processing workloads.

9.2 Reliability and Availability

- **NFR-REL-001 (Crash-Free Sessions)**: The application SHALL maintain a crash-free session rate exceeding 99.5% across all supported OS platform releases.
- **NFR-REL-002 (Data Durability)**: Meter readings, appliance inventories, and user configurations successfully committed SHALL achieve 100% durability against application termination or battery depletion.
- **NFR-REL-003 (Graceful Degradation)**: Complete unavailability of network connectivity SHALL NOT prevent users from logging readings manually, calculating consumption, or viewing local historical charts.

9.3 Security and Cryptographic Requirements

- **NFR-SEC-001 (Password Hashing)**: User passwords SHALL be hashed using Argon2id or PBKDF2 with SHA-256 and a unique per-user 128-bit cryptographic salt. Plaintext passwords SHALL NEVER touch persistent storage or application logs.
- **NFR-SEC-002 (Secure Storage at Rest)**: Local session tokens, cryptographic keys, and cached credentials SHALL be persisted exclusively within platform secure keystores (Android Keystore / iOS Keychain).
- **NFR-SEC-003 (Transport Layer Security)**: All remote network exchanges SHALL enforce TLS 1.3 with certificate pinning recommended for production endpoints.
- **NFR-SEC-004 (Sanitization & Injection Prevention)**: All manual inputs, notes, and search queries SHALL be strictly parameterized to prevent SQL injection or cross-site scripting in web views.

9.4 Privacy and Data Isolation

- **NFR-PRIV-001 (Tenant Partitioning)**: The system SHALL guarantee complete data segregation. No query executed on behalf of User A shall ever return records belonging to User B.
- **NFR-PRIV-002 (Camera & Mic Permissions)**: Hardware access SHALL be requested strictly on-demand with explicit justification dialogs. Hardware sensors SHALL be deactivated immediately upon completion of capture.

9.5 Usability and Accessibility

- **NFR-USA-001 (Touch Target Ergonomics)**: All interactive controls, buttons, and form inputs SHALL maintain a minimum physical touch target dimension of $48 \times 48\text{ dp}$ in compliance with WCAG 2.1 guidelines.
- **NFR-USA-002 (Color Contrast)**: Text, chart legends, and critical numerical indicators SHALL maintain a minimum contrast ratio of $4.5:1$ against adjacent backgrounds.
- **NFR-USA-003 (Bilingual Parity)**: Every text string, dialog, error message, and report SHALL have 100% linguistic equivalence across Egyptian Arabic and English.

9.6 Maintainability and Extensibility

- **NFR-MNT-001 (Modular Architecture)**: The software SHALL follow Clean Architecture principles (Presentation, Domain, and Data layers) to allow switching database providers without modifying business logic.

- **NFR-MNT-002 (Automated Test Coverage)**: Core domain logic (tariff calculation, electrical equations, validation rules) SHALL maintain a minimum automated unit test code coverage of *85\%*.

9.7 Portability and Compatibility

- **NFR-PORT-001 (OS Compatibility)**: The Flutter client SHALL build and execute cleanly on Android API 29+ and iOS 14.0+ from a single shared codebase.

9.8 Localization & Cultural Adaptation

- **NFR-LOC-001 (Egyptian Locale Formatting)**: Dates SHALL format as `DD/MM/YYYY` (or `YYYY/MM/DD` in Arabic context); numbers SHALL respect local decimal separators; currency SHALL display as `EGP` or `ج.م`.

9.9 Storage Architecture-Agnostic vs Dependent Matrix

To maintain rigorous separation between invariant requirements and storage-dependent constraints, the following matrix classifies non-functional behaviors:

Requirement Area	Architecture-Invariant Requirement	Storage-Dependent Requirement (Local / Cloud / Hybrid)
Data Isolation	No user can access another's data. Enforced in application domain logic.	*Cloud/Hybrid*: Enforced via database Row-Level Security (RLS) and JWT claims. *Local*: Enforced via isolated SQLite file databases.
Offline Resilience	App must allow manual reading entry and local calculations without crashing.	*Local/Hybrid*: 100% full offline read/write. *Cloud-Only*: Requires network connection or results in transient error.
Backup & Recovery	User can recover historical data upon device destruction or replacement.	*Local-Only*: Impossible without manual user export file. *Cloud/Hybrid*: Automated cloud restore upon re-authentication.
Performance	Reading lookup $< 50\text{ms}$.	*Local/Hybrid*: Read from local flash flash memory ($< 10\text{ms}$). *Cloud-Only*: Dependent on mobile cellular network latency ($150\text{ms} - 1200\text{ms}$).

10. Business Rules and Calculation Formulas

10.1 Core Business Rules Catalog

- **BR-01 (Consumption Derivation)**: Electricity consumption is derived exclusively from the difference between two valid, chronologically ordered meter readings taken on the same meter.
- **BR-02 (Chronological Monotonicity)**: For any two readings R_1 at t_1 and R_2 at t_2 where $t_2 > t_1$, the condition $R_2 \ge R_1$ MUST hold true, unless an authorized meter rollover or meter replacement event is documented.
- **BR-03 (Tariff Authority)**: The estimated monetary cost of electricity consumption is determined solely by the active tariff configuration applicable to the reading dates.
- **BR-04 (Progressive Tiering Discipline)**: In progressive electricity tariffs, consumption is partitioned into sequential tier brackets. Each bracket's consumption is multiplied by that bracket's specific unit rate, rather than multiplying total consumption by a single marginal rate.
- **BR-05 (Appliance Estimate Independence)**: Appliance-based energy figures represent bottom-up mathematical simulations based on rated specs and assumed operating schedules. They are distinct from, and SHALL NOT be conflated with, top-down actual meter readings.
- **BR-06 (AC Power Inadequacy of Voltage Alone)**: Electrical voltage (V) alone cannot determine electrical power or energy consumption. Even with current (I) known, active real power in alternating current (AC) systems depends strictly on the power factor ($\cos\phi$):

$$P (\text{Watts}) = V \times I \times \cos\phi$$

- **BR-07 (Budget Utilization Metric)**: Budget consumption percentage is calculated as:

$$\text{Budget Utilization } (\%) = \left(\frac{\text{Current Period Estimated Cost}}{\text{Configured Budget Ceiling}}\right) \times 100$$

- **BR-08 (Dual Data Stream Discrepancy)**: The system shall clearly acknowledge that meter-based consumption and appliance-aggregated estimates will rarely reconcile identically due to thermal variations, standby losses, appliance duty cycles, and unmodeled loads.
- **BR-09 (Human Confirmation Gate)**: Unconfirmed perceptual data (voice transcripts and OCR image detections) are strictly unverified candidates and SHALL NEVER be saved without explicit user verification.
- **BR-10 (Auditability & Historical Snapshotting)**: Every calculated cost estimate SHALL store an immutable snapshot of the tariff version ID and tier boundaries utilized at the moment of calculation.
- **BR-11 (Data Sovereignty)**: All private user records remain strictly confidential and partitioned to the owning user account.
- **BR-12 (Non-Deceptive Reporting)**: Missing or incomplete meter reading intervals SHALL NOT silently generate zero-cost estimates. The system SHALL visibly indicate data gaps.

10.2 Mathematical Formulations

1. Meter Consumption Delta

$$\Delta C = M_{\text{current}} - M_{\text{previous}} \quad [\text{kWh}]$$

2. Progressive Tier Cost Allocation Algorithm

Let total consumption be $C = \Delta C$. Given an ordered set of N tariff tiers $\{T_1, T_2, \dots, T_N\}$, where each tier T_i is bounded by lower limit L_i, upper limit U_i, and unit rate R_i (EGP/kWh):

The consumption allocated to tier i, denoted c_i, is defined as:

$$c_i = \max\left(0, \min\left(C, U_i\right) - L_i\right) \quad \text{for } C > L_i$$

$$\text{Cost}_{\text{tier } i} = c_i \times R_i \quad [\text{EGP}]$$

$$\text{Total Estimated Cost} = \sum_{i=1}^{N} \text{Cost}_{\text{tier } i} \quad [\text{EGP}]$$

3. AC Electrical Appliance Power and Energy

For single-phase alternating current (\text{AC}) circuits (standard Egyptian residential low-voltage grid: 220V, 50Hz):

- **Apparent Power (*S*)**:

$$S = V \times I \quad [\text{Volt-Amperes, VA}]$$

- **Active Real Power (*P*)**:

$$P = V \times I \times \cos\phi \quad [\text{Watts, W}]$$

(where \cos\phi is the circuit power factor, typically 0.85 - 0.95 for inductive motor loads like air conditioners and refrigerators, and 1.0 for purely resistive loads like heaters and incandescent lamps).

- **Periodic Energy Consumption (*E*)**:

$$E = \frac{P \times h_{\text{daily}} \times d_{\text{period}}}{1000} \times Q \quad [\text{kWh}]$$

(where h_{daily} is daily operating hours, d_{period} is operating days in period, and Q is appliance quantity).

4. Run-Rate Projected Expenditure

Let d_{elapsed} be the number of days elapsed in the current billing cycle, d_{total} be the total days in the cycle, and $\text{Cost}_{\text{elapsed}}$ be the accumulated estimated cost:

$$\text{Cost}_{\text{projected}} = \left(\frac{\text{Cost}_{\text{elapsed}}}{d_{\text{elapsed}}}\right) \times d_{\text{total}} \quad [\text{EGP}]$$

10.3 Worked Numerical Calculations (Hypothetical Egyptian Tariff)

> **NOTICE: IMPORTANT**
>
> **DISCLAIMER**: The tariff rates, tier boundaries, and customer brackets presented below are **hypothetical illustrative sample values** compiled strictly for engineering calculation verification and unit testing. They **DO NOT** represent official legal rates published by the Egyptian Ministry of Electricity and Renewable Energy or EgyptERA.

Sample Illustrative Progressive Tariff Schedule (Model: `EG_SAMPLE_RESIDENTIAL_2026`)

- **Tier 1**: $0\text{ to } 50\text{ kWh}$ @ 0.58 EGP/kWh
- **Tier 2**: $51\text{ to } 100\text{ kWh}$ @ 0.68 EGP/kWh
- **Tier 3**: $101\text{ to } 200\text{ kWh}$ @ 0.83 EGP/kWh
- **Tier 4**: $201\text{ to } 350\text{ kWh}$ @ 1.25 EGP/kWh
- **Tier 5**: $351\text{ to } 650\text{ kWh}$ @ 1.40 EGP/kWh
- **Tier 6**: $651\text{ to } 1000\text{ kWh}$ @ 1.50 EGP/kWh
- **Tier 7**: $> 1000\text{ kWh}$ @ 1.65 EGP/kWh

Worked Example 1: Standard Progressive Consumption (170 kWh)

- **Input Data**:
- Previous Meter Reading (M_{prev}): $12{,}450.0\text{ kWh}$ (Recorded 01/10/2026)
- Current Meter Reading (M_{curr}): $12{,}620.0\text{ kWh}$ (Recorded 31/10/2026)
- **Step 1: Calculate Consumption Delta**:

$$|Delta C = 12,620.0 - 12,450.0 = 170.0|text{ kWh}$$

- **Step 2: Progressive Tier Allocation**:
- **Tier 1 (*0 - 50\text{ kWh}*)**: Capacity = 50 kWh.

$$|text{Allocated} = 50.0|text{ kWh} |times 0.58|text{ EGP} = 29.00|text{ EGP}$$

$$|text{Remaining Consumption} = 170.0 - 50.0 = 120.0|text{ kWh}$$

- **Tier 2 (*51 - 100\text{ kWh}*)**: Capacity = 50 kWh.

$$|text{Allocated} = 50.0|text{ kWh} |times 0.68|text{ EGP} = 34.00|text{ EGP}$$

$$|text{Remaining Consumption} = 120.0 - 50.0 = 70.0|text{ kWh}$$

- **Tier 3 (*101 - 200\text{ kWh}*)**: Capacity = 100 kWh.

$$|text{Allocated} = 70.0|text{ kWh} |times 0.83|text{ EGP} = 58.10|text{ EGP}$$

$$|text{Remaining Consumption} = 70.0 - 70.0 = 0.0|text{ kWh}$$

- **Step 3: Total Summation**:

$$|text{Total Estimated Cost} = 29.00 + 34.00 + 58.10 = 121.10|text{ EGP}$$

Worked Example 2: Appliance Electrical Specification Calculation

- **Appliance**: Inverter Air Conditioner (1.5 HP)
- **Input Specifications**:
- Rated Voltage (V): 220 V
- Rated Current (I): 5.8 A
- Power Factor ($\cos\phi$): 0.88
- Quantity (Q): 1
- Usage: 7.5 hours/day, 30 days/month
- **Step 1: Compute Active Real Power (*P*)**:

$$P = 220 |times 5.8 |times 0.88 = 1,122.88|text{ Watts}$$

- **Step 2: Compute Monthly Energy Consumption**:

$$E = |frac{1,122.88|text{ W} |times 7.5|text{ hrs} |times 30|text{ days}}{1000} |times 1 = 252.648|text{ kWh} |approx 252.65|text{ kWh}$$

- **Step 3: Cost Evaluation**:

At an average marginal tariff rate of 1.25 EGP/kWh, the operating cost of this appliance alone is:

$$|text{Estimated Operating Cost} = 252.65 |times 1.25 = 315.81|text{ EGP/month}$$

11. Use Cases

UC-01: Register Account

- **Primary Actor**: Unauthenticated Consumer.
- **Preconditions**: Application installed, device connected to internet (if cloud auth active).
- **Trigger**: User selects "Create Account" on Welcome screen.
- **Main Success Scenario**:

1. User enters name, email, password, and password confirmation.
2. System validates email format and password strength.
3. System hashes password with Argon2id/salt.
4. System creates new `user_id` and persists user entity.
5. System generates authentication token and logs user into Dashboard.

- **Alternative Flows**:
- *4a. Email already registered*: System displays `AUTH_ERR_DUPLICATE_EMAIL`, prompts user to sign in or recover password.
- **Exception Flows**:
- *2a. Weak password*: System highlights failing complexity rules; disables submission.
- **Postconditions**: User account established; private space partitioned.
- **Related FRs**: `FR-AUTH-001`, `FR-AUTH-006`.

UC-02: Sign In

- **Primary Actor**: Registered User.
- **Preconditions**: User has an existing account.
- **Trigger**: User opens app and submits credentials.
- **Main Success Scenario**:

1. User inputs email and password, taps "Sign In".
2. System verifies credentials against stored cryptographic hash.
3. System initializes user session, loads cached profile, and opens Dashboard.

- **Alternative Flows**:
- *2a. Invalid credentials*: System increments failed attempts, displays generic error message.
- **Postconditions**: User authenticated; session token stored in secure keychain.
- **Related FRs**: `FR-AUTH-002`, `FR-AUTH-006`.

UC-03: Recover Account

- **Primary Actor**: Registered User who forgot password.
- **Trigger**: User taps "Forgot Password" on login screen.
- **Main Success Scenario**:

1. User inputs registered email address.
2. System verifies account existence; issues time-limited reset token.
3. User enters new valid password and confirms.
4. Password hash updated; previous sessions revoked.

- **Postconditions**: User password updated; access restored.
- **Related FRs**: `FR-AUTH-004`.

UC-04: Add Meter Reading Manually

- **Primary Actor**: Authenticated User.
- **Preconditions**: User has at least one registered meter.

- **Trigger**: User taps "Add Reading" on meter screen.
- **Main Success Scenario**:

1. User selects target meter.
2. User inputs numeric meter reading value and selects date/time.
3. System verifies value is *ge 0* and *ge* immediately preceding reading.
4. System saves reading with `reading_source = 'manual'`.
5. System computes consumption *Delta *text{kWh}* and estimated cost *text{EGP}*.
6. Dashboard metrics and budget utilization update immediately.

- **Alternative Flows**:
- *3a. Value lower than previous*: System warns user (`ERR_READING_BELOW_PREVIOUS`), prompts for correction or rollover confirmation (`UC-04-Alt1`).
- **Postconditions**: Reading saved; consumption calculated.
- **Related FRs**: `FR-METER-002`, `FR-METER-003`, `FR-CALC-001`.

UC-05: Enter Meter Reading Using Voice

- **Primary Actor**: Authenticated User.
- **Supporting System**: Speech Recognition Engine (OS Native or Cloud).
- **Trigger**: User taps microphone icon on reading input form.
- **Main Success Scenario**:

1. System checks/requests microphone permission.
2. System activates audio stream and visual waveform listener.
3. User speaks reading (e.g., "*Twelve thousand six hundred twenty point five*" or "*واحد اثنين ستة اثنين*").
4. Speech engine transcribes audio to text.
5. Numeric parser extracts normalized digits (*12620.5*).
6. System displays extracted number in an editable modal review dialog.
7. User reviews digits, makes optional manual adjustments, and taps "Confirm".
8. System saves reading with `reading_source = 'voice'`.

- **Exception Flows**:
- *1a. Permission denied*: System shows explanation dialog; switches focus to manual keyboard.
- *4a. Recognition failure / ambient noise*: System displays "*Voice not recognized*", prompts user to retry or type.
- **Postconditions**: Reading validated and saved with voice audit tag.
- **Related FRs**: `FR-VOICE-001` through `FR-VOICE-004`.

UC-06: Extract Meter Reading from Image (OCR)

- **Primary Actor**: Authenticated User.
- **Supporting System**: Camera Hardware & OCR Engine.
- **Trigger**: User taps "Scan Meter" button.
- **Main Success Scenario**:

1. User chooses "Take Photo" or "Choose from Gallery".
2. User captures meter image within camera framing guide.
3. System preprocesses image and runs OCR detection engine.
4. System filters text blocks to extract candidate meter readings.
5. System displays detected value in editable review modal with image crop preview.
6. User confirms accuracy and taps "Save Reading".
7. System saves reading with `reading_source = 'ocr'`.

- **Alternative Flows**:

- *4a. Multiple candidate numbers detected*: System displays disambiguation sheet showing thumbnails of detected numbers; user selects intended reading.
- **Exception Flows**:
- *3a. Blurry / Dark / Unreadable image*: System informs user: "*Could not detect clear meter digits. Please ensure good lighting and hold steady*", offers retry or manual input.
- **Postconditions**: Reading verified and persisted with OCR audit tag.
- **Related FRs**: `FR-METER-OCR-001` through `FR-METER-OCR-004`.

UC-07: Calculate Consumption and Estimated Cost

- **Primary Actor**: Application System Logic (Automated).
- **Preconditions**: At least two chronologically ordered readings exist for the meter.
- **Trigger**: Ingestion or modification of a meter reading.
- **Main Success Scenario**:

1. System queries current and immediately prior reading.
2. System computes consumption delta $\Delta C = M_{\text{curr}} - M_{\text{prev}}$.
3. System retrieves active tariff schedule covering the reading dates.
4. System executes progressive tier allocation algorithm (`BR-04`).
5. System generates consumption estimate record with immutable tier snapshot.
6. System updates budget tracker and historical analytics.

- **Postconditions**: ConsumptionEstimate record created and linked to billing period.
- **Related FRs**: `FR-CALC-001` through `FR-CALC-005`.

UC-08: View Consumption Charts

- **Primary Actor**: Authenticated User.
- **Trigger**: User navigates to "Analytics & Charts" tab.
- **Main Success Scenario**:

1. System queries user's consumption estimates and appliance inventories.
2. System renders historical consumption bar chart (Daily/Monthly/Period toggle).
3. System renders appliance distribution pie chart.
4. User taps individual bars/slices to inspect granular figures (kWh, EGP, % share).

- **Alternative Flows**:
- *1a. Insufficient historical data*: System renders helpful empty-state graphic guiding user to log further readings.
- **Postconditions**: Visual analytics presented cleanly.
- **Related FRs**: `FR-CHART-001` through `FR-CHART-004`.

UC-09: Create and Manage Budget

- **Primary Actor**: Authenticated User.
- **Trigger**: User opens Budget Management screen.
- **Main Success Scenario**:

1. User enters monthly budget limit (e.g., $500\ \text{EGP}$) and billing cycle start day.
2. System saves budget configuration.
3. System queries current period spending, computes remaining balance and % consumed.
4. If consumption exceeds 80% or 100%, system triggers warning indicator.

- **Postconditions**: Active budget monitors monthly expenditure.
- **Related FRs**: `FR-BUDGET-001` through `FR-BUDGET-004`.

UC-10: Register and Configure Appliances

- **Primary Actor**: Authenticated User.

- **Trigger**: User selects "Add Appliance".
- **Main Success Scenario**:

1. User enters appliance name, category, rated power (*W*), quantity, and daily operating hours.
2. System validates that daily hours *le 24.0*.
3. System computes monthly energy consumption (*E*) and estimated cost.
4. System saves appliance profile and links it to user's account.

- **Alternative Flows**:
- *1a. User provides Volts (V) and Amperes (A)*: System applies AC active power formula with power factor *cos*phi* (`BR-06`).
- **Postconditions**: Appliance registered and included in aggregate estimations.
- **Related FRs**: `FR-APP-001` through `FR-APP-005` .

UC-11: Compare Appliance Usage Scenarios

- **Primary Actor**: Authenticated User.
- **Trigger**: User opens "Scenario Simulator".
- **Main Success Scenario**:

1. System loads baseline appliance runtime configurations.
2. User adjusts operating hours for one or more high-consumption appliances.
3. System dynamically calculates simulated consumption and cost delta.
4. System displays side-by-side comparison: Baseline vs Simulated Scenario (*Delta *text{kWh}*, *Delta *text{EGP}* savings).

- **Postconditions**: User evaluates impact of behavioral usage modifications.
- **Related FRs**: `FR-REC-003` .

UC-12: View Usage Recommendations

- **Primary Actor**: Authenticated User.
- **Trigger**: User opens "Optimization Tips" screen.
- **Main Success Scenario**:

1. System evaluates registered appliances against active budget and consumption run-rate.
2. System identifies appliances consuming > *20\%* of domestic energy.
3. System generates tailored suggestions (e.g., "*Limiting AC usage by 1.5 hours daily will save ~45 EGP/month, keeping you within your 500 EGP budget*").

- **Postconditions**: Contextual recommendations displayed with explicit simulation disclaimer.
- **Related FRs**: `FR-REC-001` , `FR-REC-002` .

UC-13: Update or Select Applicable Tariff

- **Primary Actor**: Authenticated User / Administrator.
- **Trigger**: Regulatory tariff rate change announced.
- **Main Success Scenario**:

1. System receives or imports an updated tariff schedule JSON package.
2. System validates tier boundaries and non-negative rates.
3. System records new tariff version with effective start date without overwriting historical versions.
4. Future calculations automatically apply new schedule; historical estimates retain previous version tags.

- **Postconditions**: Tariff repository updated safely.
- **Related FRs**: `FR-TARIFF-001` through `FR-TARIFF-004` .

UC-14: Review Reading History

- **Primary Actor**: Authenticated User.
- **Trigger**: User selects "History" tab.

- **Main Success Scenario**:

1. System queries all historical readings for active user/meter ordered descending by date.
2. System displays list items showing date, reading value, delta *text{kWh}*, source tag, and notes.
3. User can filter by meter or date range, or tap an item to edit or delete.

- **Postconditions**: Audit trail displayed.
- **Related FRs**: `FR-SET-003`, `FR-METER-004`.

UC-15: Delete Account and Associated Data

- **Primary Actor**: Authenticated User.
- **Trigger**: User initiates account deletion in Settings.
- **Main Success Scenario**:

1. System requests password re-entry and displays explicit warning of irreversible data destruction.
2. User re-enters password and confirms.
3. System executes cascading deletion across all user records in local database (and remote backend).
4. Local credentials and tokens purged.
5. User navigated to initial Welcome screen.

- **Postconditions**: User identity and all private historical records permanently eradicated.
- **Related FRs**: `FR-AUTH-005`, `FR-PRIV-001`.

12. Data Requirements and Logical Data Model

12.1 Entity Catalog & Dictionary

The logical data model comprises 13 distinct entities engineered independently of specific database vendor implementations:

#	Entity Name	Business Purpose
1	USER	Authenticated consumer identity, credentials & audit timestamps
2	METER	Physical or virtual electricity meter profile
3	METER_READING	Recorded meter reading entry with timestamp and source audit
4	TARIFF_CONFIGURATION	Master tariff version header and effective date range
5	TARIFF_TIER	Individual progressive tier bracket bounds and rate per kWh
6	BILLING_PERIOD	Calendar or custom billing cycle interval
7	BUDGET	Financial ceiling and warning thresholds for a billing period
8	APPLIANCE	Electrical appliance technical specifications and inventory
9	APPLIANCE_USAGE_SCHEDULE	Operating hours and periodic duty cycle for an appliance
10	CONSUMPTION_ESTIMATE	Calculated kWh delta, estimated cost in EGP, and tier snapshot
11	USAGE_SCENARIO	Simulated what-if behavioral usage scenario configuration
12	RECOMMENDATION	Optimization guidance generated by rule engine
13	APPLICATION_SETTINGS	User preferences (language, currency format, theme)

Entity 1: USER

- `user_id` (String, PK, UUID v4): Unique system identifier.
- `email` (String, UK, Not Null): RFC 5322 formatted email address.
- `password_hash` (String, Not Null): Argon2id/PBKDF2 salted password hash.
- `display_name` (String, Optional): User's preferred name.
- `is_verified` (Boolean, Default: False): Email verification flag.
- `created_at` (Timestamp, Not Null): Account creation timestamp.
- `updated_at` (Timestamp, Not Null): Last profile update timestamp.

Entity 2: METER

- `meter_id` (String, PK, UUID v4): Unique meter identifier.
- `user_id` (String, FK -> USER.user_id, Not Null): Owning user reference.
- `meter_name` (String, Not Null): User-defined alias (e.g., "Apartment 4B").
- `meter_serial_number` (String, Optional): Manufacturer physical serial.
- `meter_type` (Enum: `MECHANICAL`, `DIGITAL_LCD`, `PREPAID_SMART`): Meter hardware classification.
- `created_at` (Timestamp, Not Null).

Entity 3: METER_READING

- `reading_id` (String, PK, UUID v4): Unique reading record identifier.
- `meter_id` (String, FK -> METER.meter_id, Not Null): Meter reference.
- `user_id` (String, FK -> USER.user_id, Not Null): Security isolation owner.
- `reading_value` (Double, Not Null): Numeric reading in *text{kWh}* (*ge 0*).
- `reading_timestamp` (Timestamp, Not Null): Moment reading was taken.
- `reading_source` (Enum: `MANUAL`, `VOICE`, `OCR`): Ingestion modality.
- `notes` (String, Optional): Contextual user observations.
- `created_at` (Timestamp, Not Null).

Entity 4: TARIFF_CONFIGURATION

- `tariff_id` (String, PK): Unique tariff schedule identifier.
- `version_name` (String, Not Null): Descriptive code (e.g., "EG_RES_2026_V1").
- `effective_from` (Date, Not Null): Activation date.
- `effective_to` (Date, Optional): Expiration date (null if current).
- `customer_category` (Enum: `RESIDENTIAL`, `COMMERCIAL`).
- `calculation_model` (Enum: `PROGRESSIVE_BRACKET`, `SLAB_RESET`).
- `is_active` (Boolean, Not Null).

Entity 5: TARIFF_TIER

- `tier_id` (String, PK): Unique tier bracket identifier.
- `tariff_id` (String, FK -> TARIFF_CONFIGURATION.tariff_id, Not Null): Parent schedule.
- `tier_order` (Integer, Not Null): Sequence index ($1, 2, 3 \dots$).
- `tier_start_kwh` (Double, Not Null): Lower bound (≥ 0).
- `tier_end_kwh` (Double, Not Null): Upper bound (∞ for top tier).
- `rate_per_kwh` (Double, Not Null): Unit price in EGP/kWh.
- `tier_label` (String, Not Null): Localized descriptor (e.g., "Tier 1: 0-50 kWh").

Entity 6: BILLING_PERIOD

- `period_id` (String, PK, UUID v4): Unique cycle identifier.
- `user_id` (String, FK -> USER.user_id, Not Null): Owning user.
- `start_date` (Date, Not Null): Cycle start date.
- `end_date` (Date, Not Null): Cycle end date.
- `status` (Enum: `OPEN`, `CLOSED`).

Entity 7: BUDGET

- `budget_id` (String, PK, UUID v4): Unique budget record identifier.
- `user_id` (String, FK -> USER.user_id, Not Null): Owning user.
- `period_id` (String, FK -> BILLING_PERIOD.period_id, Not Null): Associated cycle.
- `budget_amount_egp` (Double, Not Null): Spending ceiling in EGP (> 0).
- `warning_threshold_pct` (Double, Default: 80.0): Early-alert threshold.
- `created_at` (Timestamp, Not Null).

Entity 8: APPLIANCE

- `appliance_id` (String, PK, UUID v4): Unique appliance record identifier.
- `user_id` (String, FK -> USER.user_id, Not Null): Owning user.
- `name` (String, Not Null): Descriptive label (e.g., "Kitchen Refrigerator").
- `category` (Enum: `COOLING`, `HEATING`, `REFRIGERATION`, `LIGHTING`, `ENTERTAINMENT`, `COOKING`, `PUMP`, `OTHER`).
- `rated_power_watts` (Double, Not Null): Active power in Watts (> 0).
- `rated_voltage_volts` (Double, Optional): Operational voltage (V).
- `rated_current_amperes` (Double, Optional): Operational current (A).
- `power_factor` (Double, Default: 0.90): Dimensionless ratio ($0.0 < \cos\phi \le 1.0$).
- `quantity` (Integer, Default: 1): Count of identical units (≥ 1).
- `standby_power_watts` (Double, Default: 0.0).
- `created_at` (Timestamp, Not Null).

Entity 9: APPLIANCE_USAGE_SCHEDULE

- `schedule_id` (String, PK, UUID v4): Unique schedule identifier.
- `appliance_id` (String, FK -> APPLIANCE.appliance_id, Not Null): Parent appliance.
- `daily_operating_hours` (Double, Not Null): Operating hours per day ($0.0 \le h \le 24.0$).
- `operating_days_per_month` (Integer, Default: *30*): Operating days per cycle ($1 \le d \le 31$).
- `usage_pattern` (Enum: `CONTINUOUS`, `INTERMITTENT`, `PEAK_ONLY`).

Entity 10: CONSUMPTION_ESTIMATE

- `estimate_id` (String, PK, UUID v4): Unique estimate identifier.
- `user_id` (String, FK -> USER.user_id, Not Null): Owning user.
- `meter_id` (String, FK -> METER.meter_id, Not Null): Meter reference.
- `current_reading_id` (String, FK -> METER_READING.reading_id, Not Null).
- `previous_reading_id` (String, FK -> METER_READING.reading_id, Not Null).
- `tariff_id` (String, FK -> TARIFF_CONFIGURATION.tariff_id, Not Null).
- `period_id` (String, FK -> BILLING_PERIOD.period_id, Optional).
- `consumption_kwh` (Double, Not Null): Delta usage in kWh.
- `estimated_cost_egp` (Double, Not Null): Computed monetary cost.
- `calculation_snapshot_json` (String, Not Null): Complete JSON audit record of tier breakdown and rates applied.
- `calculated_at` (Timestamp, Not Null).

Entity 11: USAGE_SCENARIO

- `scenario_id` (String, PK, UUID v4): Unique scenario identifier.
- `user_id` (String, FK -> USER.user_id, Not Null): Owning user.
- `scenario_name` (String, Not Null): Alias (e.g., "Summer Energy Saving Plan").
- `baseline_snapshot_json` (String, Not Null): Pre-simulation appliance state.
- `simulated_adjustments_json` (String, Not Null): Modified runtimes applied.
- `projected_kwh` (Double, Not Null): Simulated aggregate usage.
- `projected_cost_egp` (Double, Not Null): Simulated estimated cost.
- `estimated_savings_egp` (Double, Not Null): Cost delta vs baseline.
- `created_at` (Timestamp, Not Null).

Entity 12: RECOMMENDATION

- `recommendation_id` (String, PK, UUID v4): Unique recommendation identifier.
- `user_id` (String, FK -> USER.user_id, Not Null): Owning user.
- `appliance_id` (String, FK -> APPLIANCE.appliance_id, Optional).
- `scenario_id` (String, FK -> USAGE_SCENARIO.scenario_id, Optional).
- `title` (String, Not Null): Brief recommendation headline.
- `description` (String, Not Null): Contextual explanation of suggested action.
- `potential_kwh_reduction` (Double, Not Null): Projected energy reduction.
- `potential_cost_savings_egp` (Double, Not Null): Projected monetary reduction.
- `generated_at` (Timestamp, Not Null).

Entity 13: APPLICATION_SETTINGS

- `settings_id` (String, PK, UUID v4): Unique settings record identifier.
- `user_id` (String, FK -> USER.user_id, UK, Not Null): 1-to-1 link to user.
- `language_code` (Enum: `ar`, `en`, Default: `ar`): Interface locale.
- `currency_code` (String, Default: "EGP"): Display currency.

- `digit_format` (Enum: `WESTERN_0_9`, `EASTERN_ARABIC`): Numeral display mode.
- `default_warning_threshold` (Double, Default: *80.0*): Default budget alert %.
- `theme_mode` (Enum: `LIGHT`, `DARK`, `SYSTEM`).
- `updated_at` (Timestamp, Not Null).

12.2 Calculated vs Persisted Data Entities

To balance query performance against data storage integrity, the logical model strictly demarcates persisted vs dynamically computed values:

Data Value	Storage Strategy	Technical & Business Justification
Meter Reading Value (*\text{kWh}***)**	Persisted (`METER_READING`)	Raw historical fact; cannot be derived.
Consumption Delta (*\Delta \text{kWh}***)**	Persisted (`CONSUMPTION_ESTIMATE`)	Preserves immutable calculation baseline even if subsequent readings mutate.
Estimated Cost (*\text{EGP}***)**	Persisted (`CONSUMPTION_ESTIMATE`)	Must remain historically frozen to reflect rates at the time of calculation.
Tariff Breakdown Snapshot	Persisted JSON Snapshot	Critical for auditability; explains historical bills if official tariffs change.
Budget Remaining Balance	Calculated Dynamically at Runtime	Derived as *\text{Budget} - \sum(\text{Active Period Costs})* to reflect real-time state.
Budget Run-Rate Projection	Calculated Dynamically at Runtime	Depends on daily progress within active billing cycle.
Appliance Monthly Consumption	Calculated Dynamically at Runtime	Directly computed from wattage and scheduled operating hours.

12.3 Data Integrity & Ownership Rules

1. **Cascade Deletion**: Deleting a `USER` SHALL automatically cascade and purge all dependent child records (`METER`, `METER_READING`, `CONSUMPTION_ESTIMATE`, `BUDGET`, `APPLIANCE`, `USAGE_SCENARIO`, `APPLICATION_SETTINGS`).
2. **Foreign Key Integrity**: A `METER_READING` cannot exist without an active parent `METER`.
3. **Immutability of Historical Snapshots**: Once committed, the `calculation_snapshot_json` in `CONSUMPTION_ESTIMATE` SHALL NOT be updated; adjustments require generating a new estimate record.

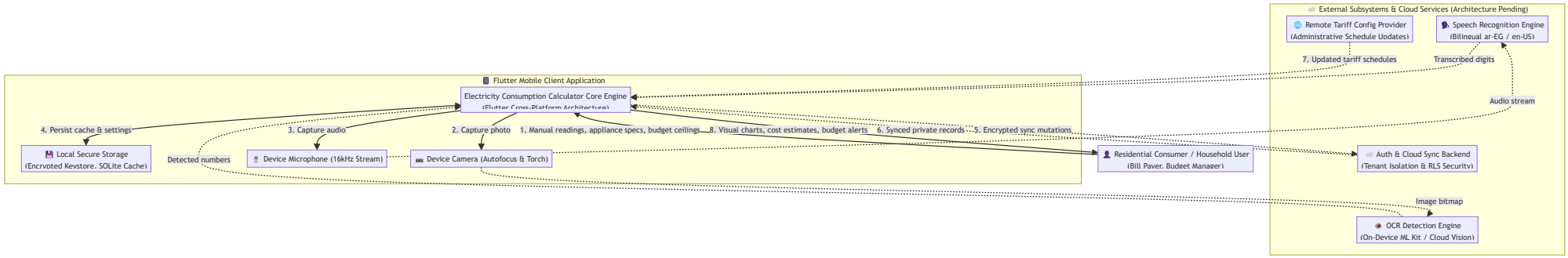
13. System Diagrams

All 9 required system diagrams have been authored in standardized Mermaid notation. Editable source files are stored in the project's `diagrams/` directory.

13.1 System Context Diagram

Source File: `diagrams/system_context.mmd`

Figure 13.1: System Context Diagram

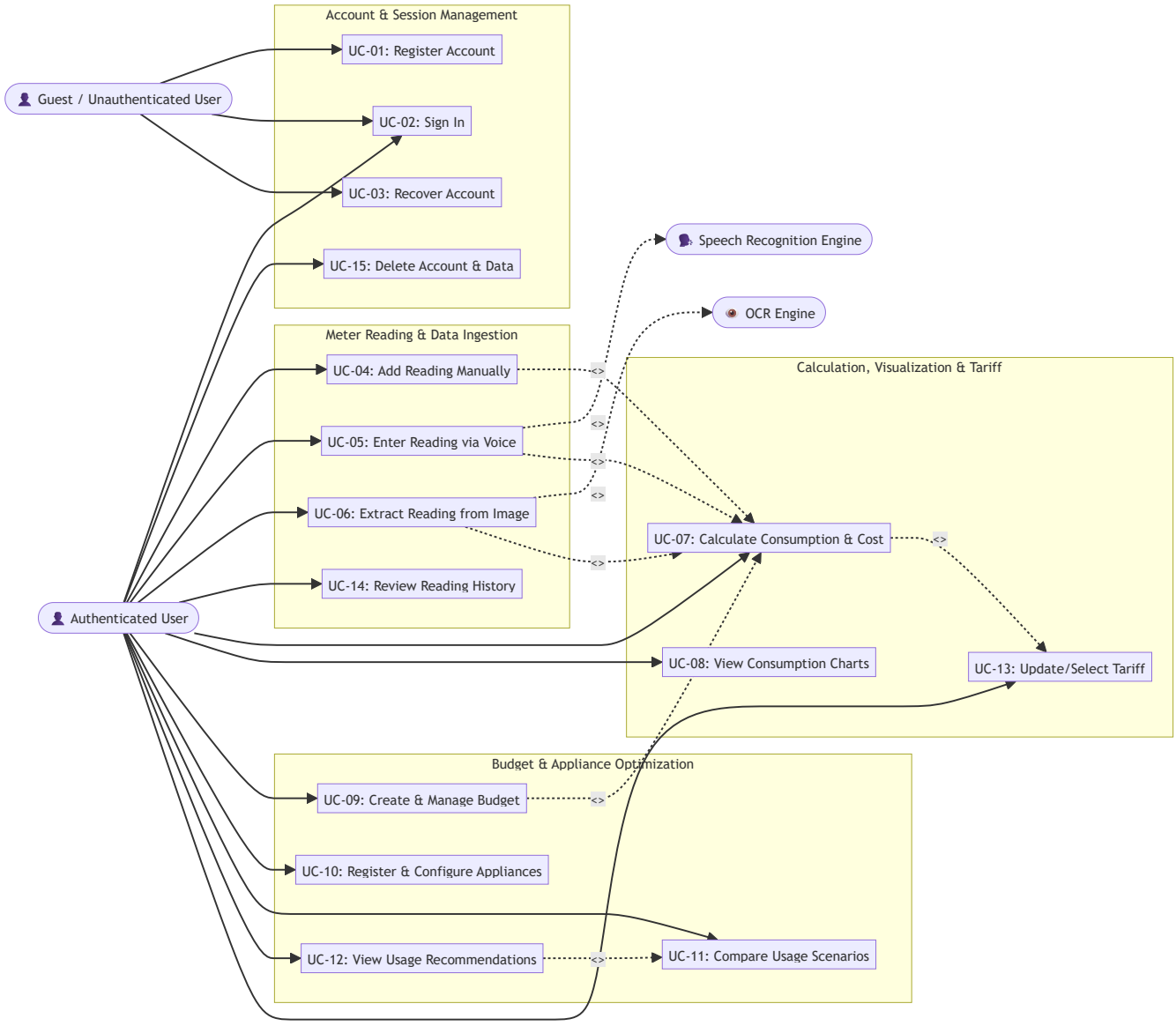


Illustrates high-level system boundary, human actors, local device hardware peripherals, and external cloud services.

13.2 Use Case Diagram

Source File: `diagrams/use_case.mmd`

Figure 13.2: Formal Use Case Diagram (UC-01 through UC-15)

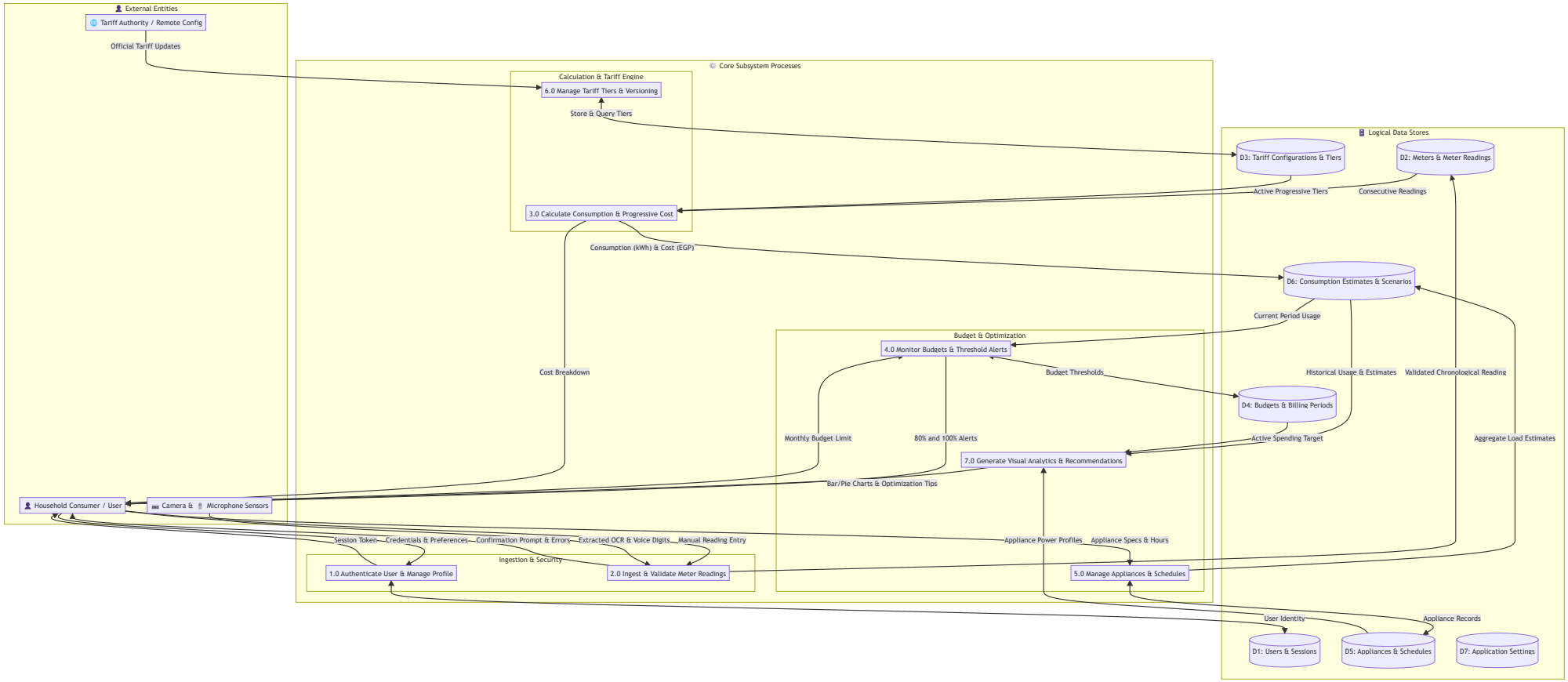


Depicts all 15 functional use cases categorized into 4 core subsystems with include/uses associations.

13.3 Level 0 Data Flow Diagram (DFD-0)

Source File: `diagrams/dfd_level_0.mmd`

Figure 13.3: Level 0 Data Flow Diagram (DFD-0)

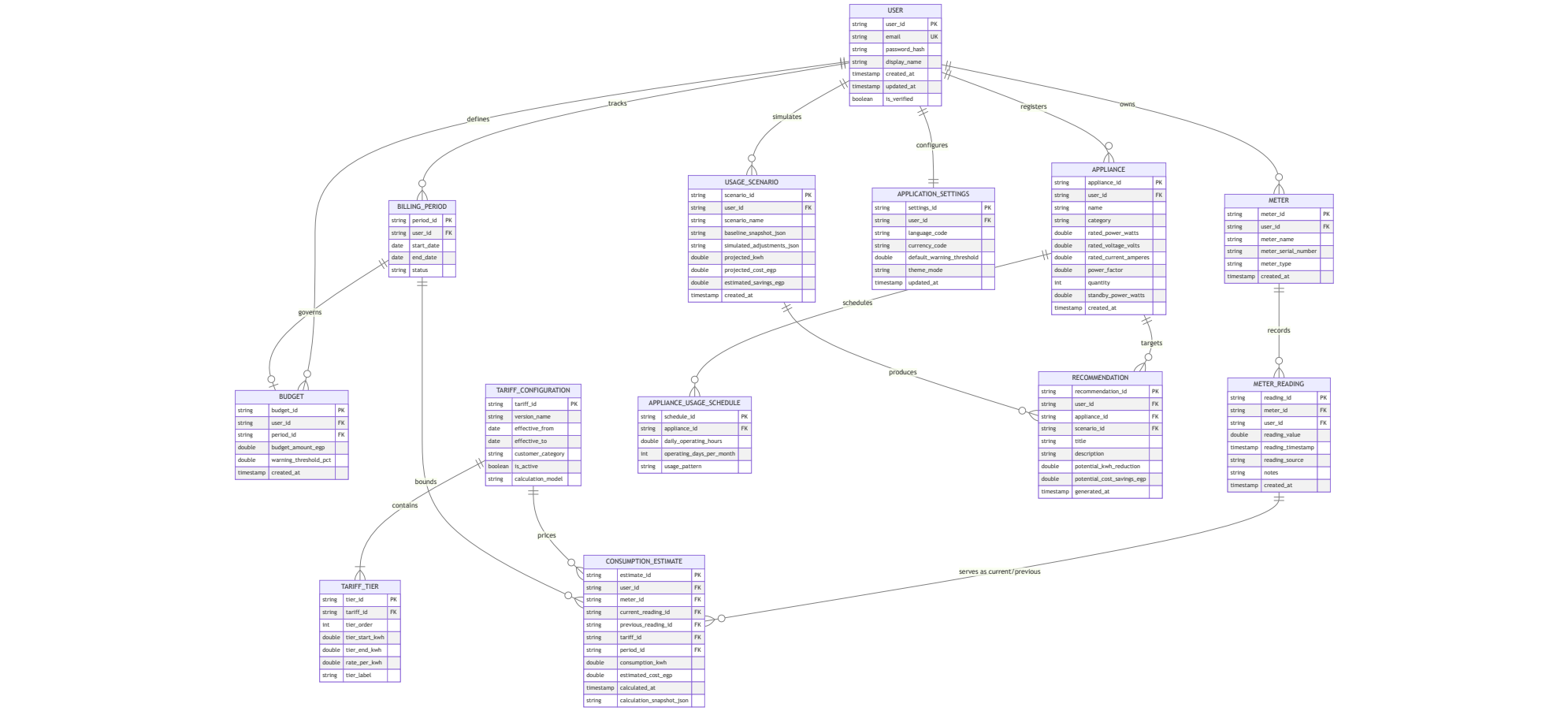


Maps data transformation processes 1.0 to 7.0, data stores D1 to D7, and external sensor data streams.

13.4 Entity-Relationship Diagram (ERD)

Source File: `diagrams/erd.mmd`

Figure 13.4: Complete Entity-Relationship Diagram (13 Entities)

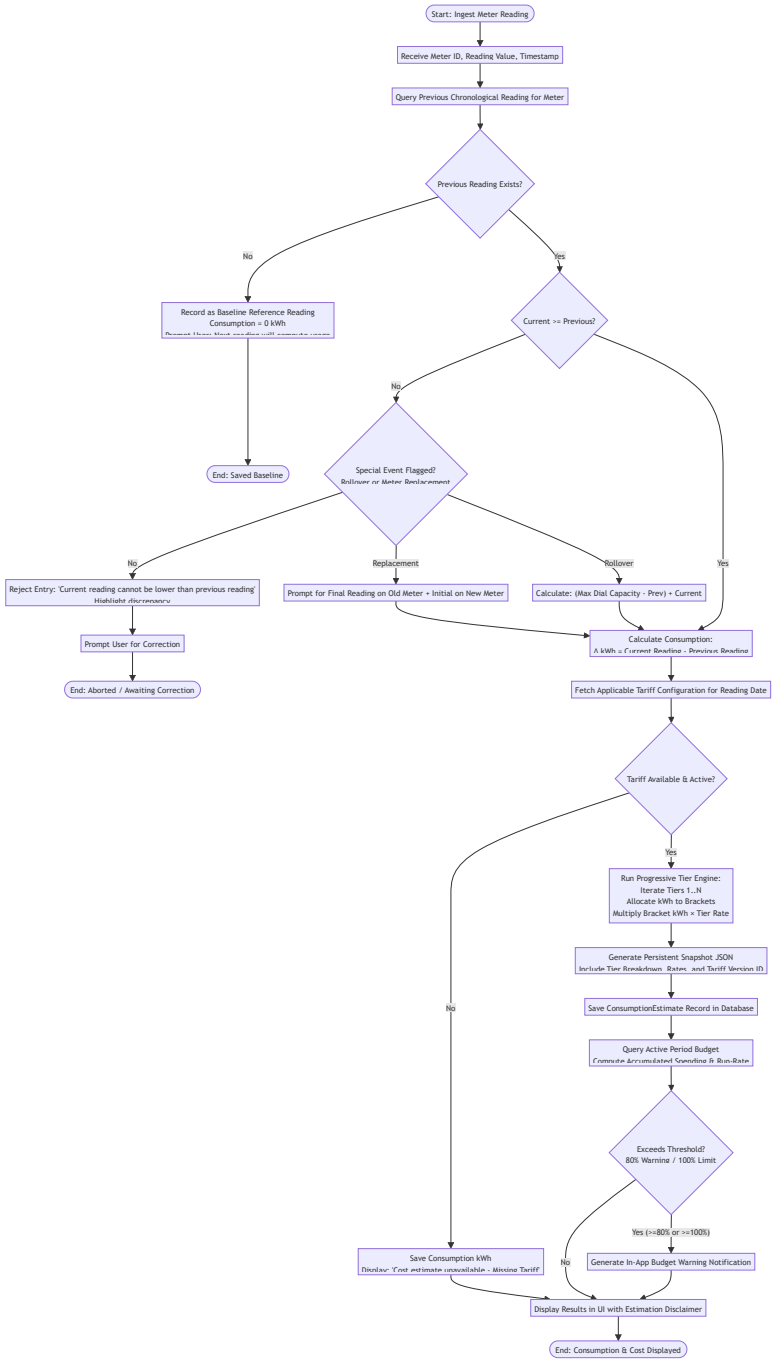


Defines complete relational data schema, primary keys, foreign keys, and cardinalities.

13.5 Activity Diagram: Meter-Based Consumption Calculation

Source File: `diagrams/activity_meter_calc.mmd`

Figure 13.5: Activity Diagram - Meter Consumption & Progressive Cost Calculation

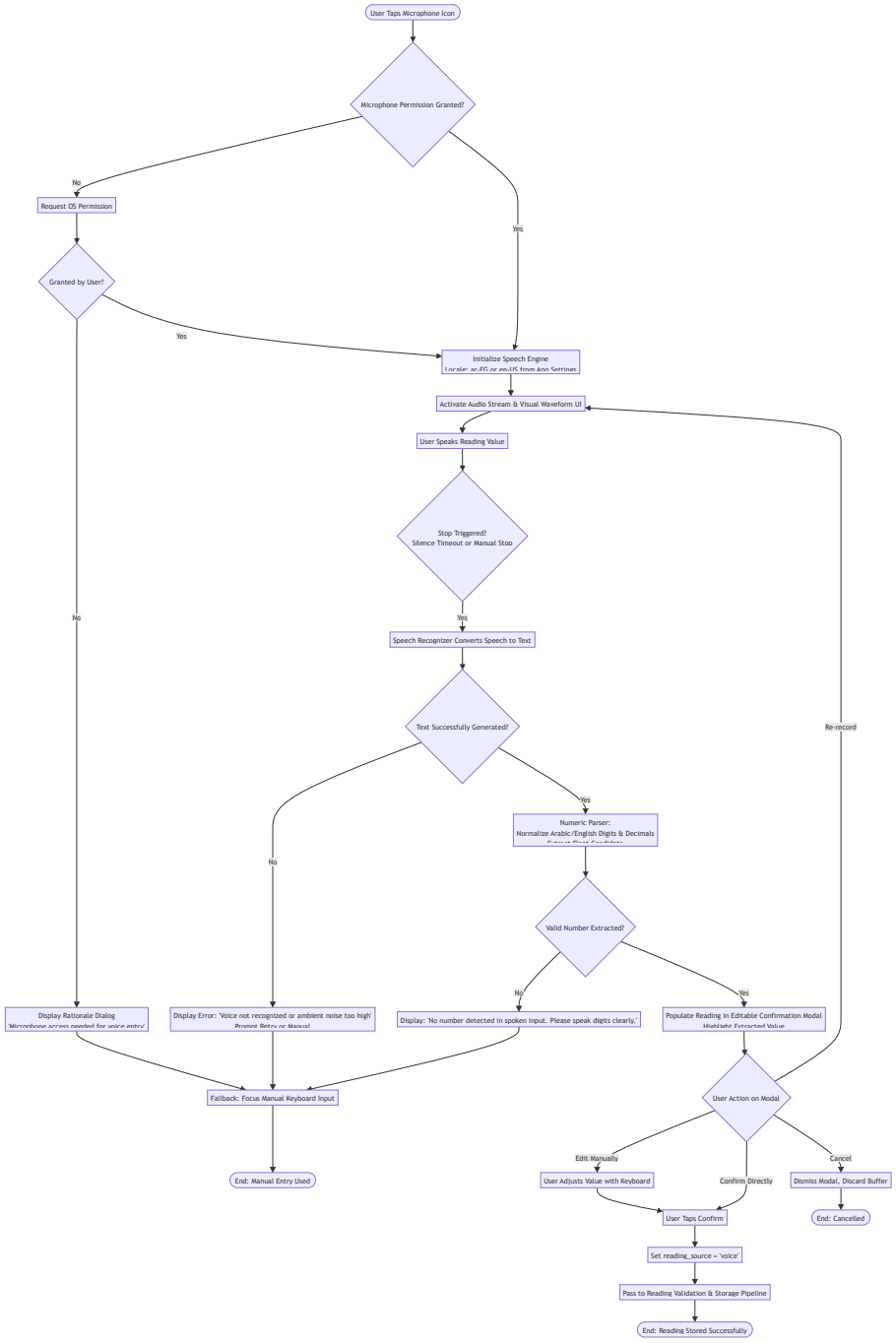


Workflow covering chronological validation, rollover handling, progressive tier allocation, and budget alerts.

13.6 Activity Diagram: Voice-Based Reading Entry

Source File: `diagrams/activity_voice_entry.mmd`

Figure 13.6: Activity Diagram - Voice-Based Meter Reading Entry

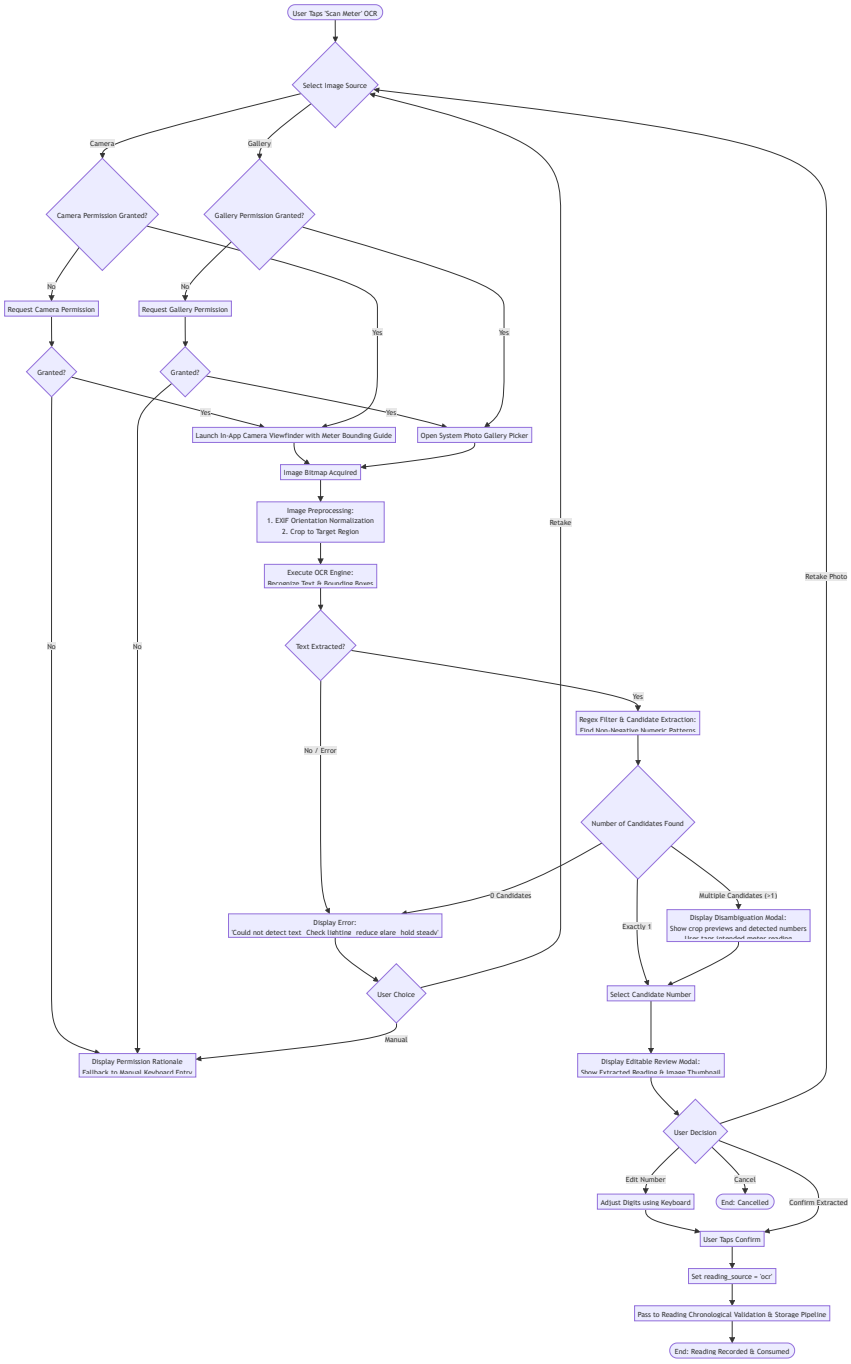


Workflow covering mic permissions, bilingual speech recognition, numeric parsing, and confirmation gate.

13.7 Activity Diagram: OCR-Based Reading Entry

Source File: `diagrams/activity_ocr_entry.mmd`

Figure 13.7: Activity Diagram - OCR-Based Meter Reading Extraction

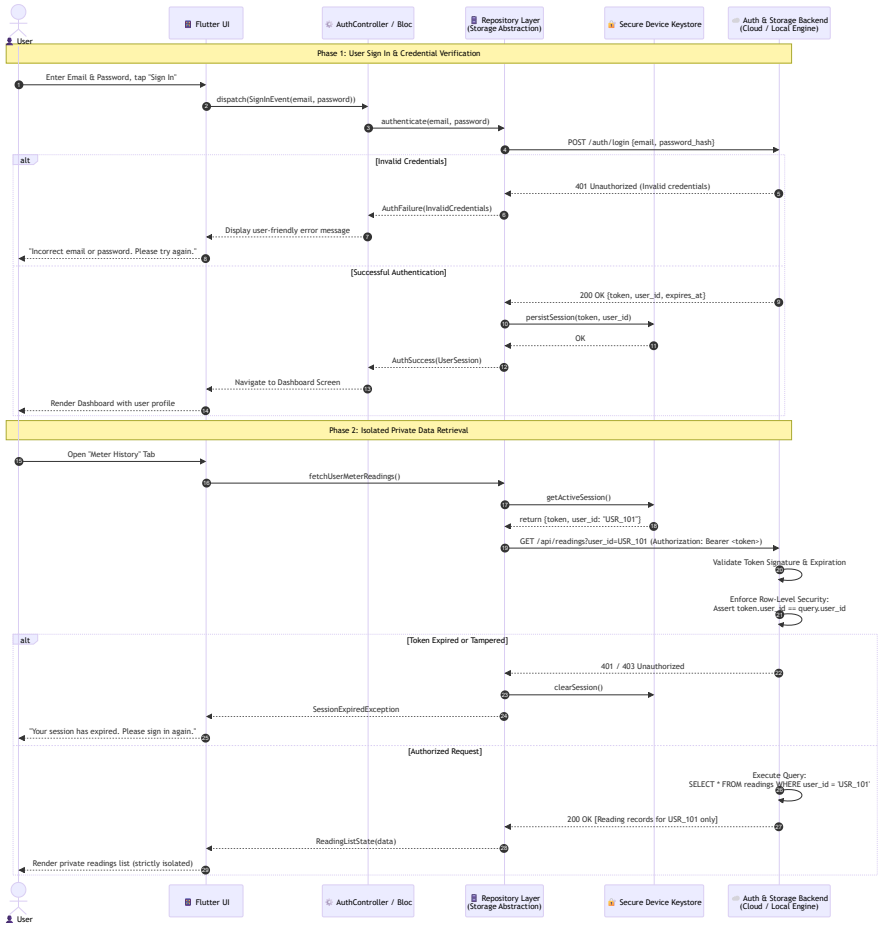


Workflow covering image capture, preprocessing, text detection, multi-candidate disambiguation, and verification.

13.8 Sequence Diagram: Authentication and Private Data Access

Source File: `diagrams/sequence_auth_data.mmd`

Figure 13.8: Sequence Diagram - User Authentication & Private Data Access

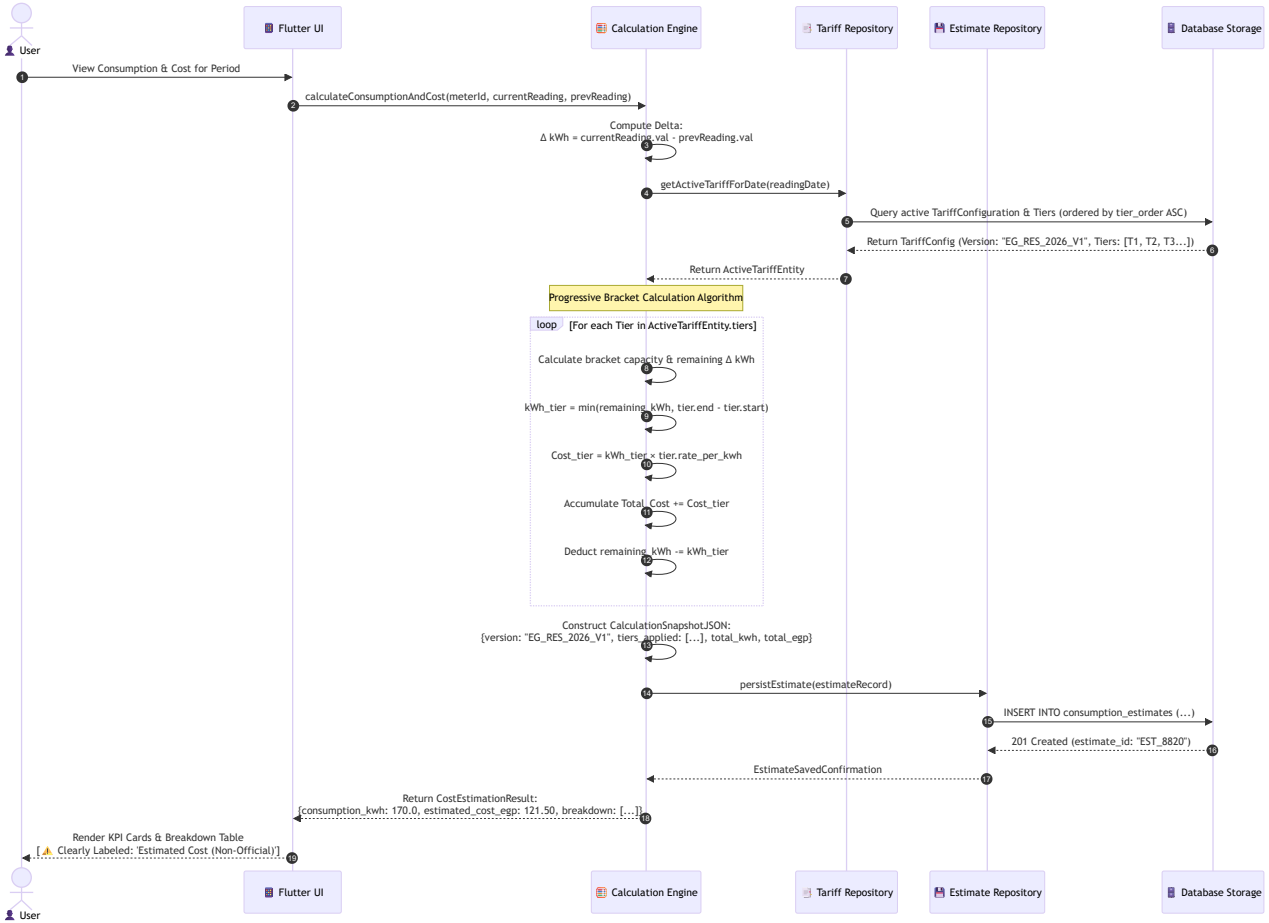


Interactions across Presentation, Auth Controller, Storage Repository, and Secure Keystore enforcing tenant isolation.

13.9 Sequence Diagram: Tariff-Based Cost Estimation

Source File: `diagrams/sequence_tariff_cost.mmd`

Figure 13.9: Sequence Diagram - Tariff-Based Cost Estimation



Interactions across UI, Calculation Engine, Tariff Repository, and Estimate Storage for progressive cost snapshotting.

14. Error Handling and Edge Cases

The following comprehensive matrix specifies failure scenarios, diagnostic detection mechanisms, user-facing notifications, recovery procedures, and internal system fallbacks across all 22 identified edge cases:

#	Edge Case / Condition	Trigger / Cause	System Detection	User-Facing Error Message	Recovery Action	System Fallback
EC-01	Negative Meter Reading Value	User enters negative number (e.g., `-150`).	Form input validation filter: `value < 0`.	*"Meter reading must be a positive number."*	Reject input; highlight field in red.	Prevent record submission.
EC-02	Current Reading Below Previous Reading	User inputs reading lower than preceding date without rollover tag.	Chronological check: $R_{\text{curr}} < R_{\text{prev}}$.	*"Current reading cannot be lower than the previous reading (12,450 kWh). If your meter rolled over or was replaced, select the appropriate option."*	Prompt user to verify reading or check "Meter Replacement / Rollover".	Abort save; preserve previous valid reading state.
EC-03	Duplicate Reading Timestamp	Submitting reading with identical date/time on same meter.	Unique constraint check on `(meter_id, reading_timestamp)`.	*"A reading for this exact date and time already exists."*	Prompt user to edit existing reading or modify timestamp.	Reject insertion.
EC-04	Chronological Date Inversion	Setting a reading date in the past that contradicts adjacent readings.	Inter-reading boundary assertion: $R_{t-1} \le R_t \le R_{t+1}$.	*"The date selected places this reading between existing readings with conflicting values."*	Guide user to view reading history and select an aligned date.	Abort insertion.
EC-05	Missing Historical Reading (Gap)	Significant multi-month gap between readings.	Days elapsed between consecutive readings > 60 days.	*"Notice: 68 days elapsed since your last reading. Consumption will be averaged over this multi-month period."*	Informational banner; allow save.	Distribute consumption proportionally across elapsed billing cycles.
EC-06	Malformed Numeric String	Special characters, multiple decimal points, or letters in field.	Regex validation failure (`^\d+(\.\d{1,2})?$`).	*"Please enter a valid numeric reading (e.g., 12450.5)."*	Sanitize input; block submission until valid.	Set field focus to error location.

#	Edge Case / Condition	Trigger / Cause	System Detection	User-Facing Error Message	Recovery Action	System Fallback
EC-07	Voice Recognition Failure	Ambient background noise, silence, or low confidence.	Speech recognizer returns null string, timeout, or confidence score < *0.5*.	"*Could not understand audio. Please speak clearly near the microphone or enter the reading manually.*"	Prompt retry with waveform indicator or direct to keyboard.	Revert focus to manual text input.
EC-08	Microphone Permission Denied	User denies audio recording OS permission.	Platform permission status `PermissionStatus.denied`.	"*Microphone permission is required for voice entry. You can enable it in system settings or type the reading manually.*"	Display setting shortcut or switch to keyboard input.	Disable microphone button; default to manual entry.
EC-09	OCR Text Extraction Failure	Blurry photo, extreme glare on meter glass, dark room.	OCR engine returns 0 text blocks or 0 numeric candidates.	"*No meter reading could be detected. Please ensure good lighting, avoid reflections, and hold the camera steady.*"	Display "Retake Photo" and "Enter Manually" buttons.	Retain captured image in memory until user decides.
EC-10	Multiple OCR Numeric Candidates	Image captures meter serial number and dial digits.	Candidate filter returns > *1* plausible numeric strings.	"*Multiple numbers detected. Please tap the correct meter reading below.*"	Display crop thumbnails of detected candidates for user selection.	Do not automatically guess; require human selection.
EC-11	Missing Active Tariff Configuration	System date falls outside all configured tariff validity ranges.	Tariff repository returns null for target reading date.	"*No active electricity tariff found for this period. Consumption calculated in kWh, but cost estimates are unavailable.*"	Prompt user/admin to update tariff schedule in Settings.	Save *\text{kWh}* consumption; store `estimated_cost_egp = NULL`.
EC-12	Mid-Period Tariff Rate Change	Government updates rates halfway through a user's billing cycle.	Billing period intersects two distinct tariff `effective_from` dates.	"*Notice: Electricity rates changed during this period. Consumption has been calculated using weighted pro-rata tier rates.*"	Display calculation details showing weighted split.	Execute pro-rata day-weighted cost calculation.

#	Edge Case / Condition	Trigger / Cause	System Detection	User-Facing Error Message	Recovery Action	System Fallback
EC-13	Invalid / Overlapping Tariff Tiers	Corrupt tariff JSON containing negative rates or tier gaps.	Tariff validator detects $U_i \neq L_{i+1}$ or $R_i < 0$.	*"Tariff configuration error: Invalid tier boundaries detected. Please restore default tariff schedule."*	Offer "Reset to Standard Tariff" button in Settings.	Fallback to immutable baseline factory default tariff schedule.
EC-14	Budget Exceeded ($100\%+$)	Estimated monthly spending surpasses user budget ceiling.	Evaluation: $\text{Cost to Date} \geq \text{Budget Amount}$.	*"⚠️ Budget Alert: You have exceeded your 500 EGP budget by 45.20 EGP."*	Highlight budget card in red; offer usage optimization suggestions.	Log alert in local notification center; no blocking of calculations.
EC-15	Excessive Appliance Operating Hours	User enters $> 24.0 \text{ hours/day}$ for an appliance.	Validation: `daily_operating_hours > 24.0`.	*"Operating hours cannot exceed 24 hours per day."*	Reset field to *24.0*; display validation tooltip.	Block appliance save action.
EC-16	Missing Appliance Wattage	User provides Volts and Amperes but omits Watts and $\cos\phi$.	Field check: `power_watts == null && (volts != null && amps != null)`.	*"Rated power calculated assuming standard power factor (0.90). You can adjust this in advanced settings."*	Informational banner; compute $P = V \times I \times 0.90$.	Populate calculated wattage into active power field.
EC-17	External Service Unavailability	Network loss while invoking cloud STT/OCR or sync backend.	HTTP timeout ($> 5000 \text{ ms}$) or `SocketException`.	*"Network connection unavailable. Using offline manual mode."*	Transition gracefully to local manual workflow.	Maintain local offline cache queue.
EC-18	Expired Authentication Session	Session token expires or is revoked.	Storage backend returns `401 Unauthorized`.	*"Your session has expired. Please sign in again to access your records."*	Redirect to login screen; preserve pending uncommitted draft reading.	Clear expired token from secure storage.
EC-19	Unauthorized Data Access Attempt	Manipulated query attempting to read another `user_id`.	Row-Level Security violation / ID mismatch assertion.	*"Access Denied: You do not have permission to view these records."*	Abort request; log security anomaly.	Return empty result set / terminate session.
EC-20	Data Synchronization Conflict	Same reading modified independently on two offline devices.	Cloud sync detects conflicting `updated_at` timestamps.	*"A sync conflict occurred for reading on [Date]. Which version would you like to keep?"*	Display side-by-side diff modal (Device A vs Cloud) for user resolution.	Default to "Last Write Wins" timestamp if unresolved.
EC-21	Local Database Storage Full / IO Failure	Device flash storage completely exhausted (0 bytes free).	SQLite throws `SQLITE_FULL` or file write exception.	*"Storage full: Cannot save reading. Please free up space on your device."*	Alert user; prevent silent data loss.	Hold uncommitted reading in volatile RAM memory.

#	Edge Case / Condition	Trigger / Cause	System Detection	User-Facing Error Message	Recovery Action	System Fallback
EC-22	Cascading Account Deletion Failure	Network drops halfway through remote account purge.	Deletion transaction fails or times out.	*"Account deletion could not be completed due to a connection error. Please try again."*	Rollback partial local deletions; prompt user to retry.	Maintain transactional rollback integrity.

15. Acceptance Criteria and Test Specifications

> **NOTICE: NOTE**
>
> The test cases cataloged below represent **formal verification specifications** designed for QA automation engineers and acceptance testing. They are defined specifications and do not imply prior manual test execution.

15.1 Positive Functional Tests

TC-AUTH-001: Valid User Registration and Initial State

- **Preconditions**: Application freshly installed; test database clean.
- **Test Steps**:

1. Open application and tap "Register".
2. Enter Name: "Ahmed Ali", Email: `ahmed.test@example.com`, Password: `Password@123`, Confirm: `Password@123`.
3. Submit registration form.

- **Expected Results**:
- Registration succeeds in $< 1.5\text{ seconds}$.
- User record created with unique UUID `user_id`.
- Password stored as salted Argon2id/PBKDF2 hash (plaintext string `Password@123` does not appear anywhere in storage).
- Default `APPLICATION_SETTINGS` record instantiated (Language: `ar`, Currency: `EGP`).
- User redirected to empty Dashboard with welcome onboarding prompt.

TC-METER-001: Successive Meter Reading Ingestion & Delta Calculation

- **Preconditions**: Authenticated user `USR_01` has active meter `MET_01`.
- **Test Steps**:

1. Add Reading 1: Value = $10{,}000.0\text{ kWh}$, Date = $01/09/2026\text{ 10:00}$.
2. Verify Baseline State: Consumption = 0.0 kWh, Status: "Baseline Established".
3. Add Reading 2: Value = $10{,}250.0\text{ kWh}$, Date = $01/10/2026\text{ 10:00}$.

- **Expected Results**:
- Reading 2 successfully saved with `reading_source = 'manual'`.
- Calculated consumption $\Delta C = 10{,}250.0 - 10{,}000.0 = 250.0\text{ kWh}$.
- Elapsed duration calculated as exactly 30 days.
- Daily average consumption displayed as 8.33 kWh/day.

15.2 Negative & Boundary Value Tests

TC-METER-002: Boundary Test — Lower Reading Value Rejection

- **Preconditions**: Existing reading on 15/10/2026 is $5{,}400.0\text{ kWh}$.
- **Test Steps**:

1. Attempt to add new reading on 20/10/2026 with value $5{,}399.9\text{ kWh}$.
2. Submit form without checking "Rollover / Replacement".

- **Expected Results**:
- Form submission is blocked.
- UI displays error: `ERR_READING_BELOW_PREVIOUS` ("*Current reading cannot be lower than previous reading*").
- No database insert occurs.

TC-APP-001: Boundary Test — Maximum Daily Operating Hours

- **Preconditions**: Appliance creation form open.
- **Test Steps**:

1. Enter Name: "Water Heater", Power: *1500\text{ W}*.
2. Enter Daily Operating Hours: *24.1\text{ hours}*.
3. Attempt to save.

- **Expected Results**:
- Validation error triggered: "*Operating hours cannot exceed 24 hours per day*".
- Field value constrained or clamped to *24.0*.

15.3 Security and User Isolation Tests

TC-SEC-001: Cross-Tenant Data Isolation Verification

- **Preconditions**: User A (`USR_A`) has 5 meter readings. User B (`USR_B`) logs in.
- **Test Steps**:

1. Authenticate as `USR_B` .
2. Query meter reading history list.
3. Attempt direct repository query requesting `meter_id` belonging to `USR_A` .

- **Expected Results**:
- UI renders zero readings for `USR_B` .
- Repository layer asserts `user_id == current_authenticated_user_id` and rejects cross-tenant query with `SecurityException` or empty set.
- No leakage of `USR_A` 's consumption or appliance data.

15.4 Mathematical & Progressive Tariff Calculation Tests

TC-CALC-001: Progressive Tier Cost Calculation Verification

- **Test Configuration**: Apply hypothetical sample tariff `EG_SAMPLE_RESIDENTIAL_2026` (Tiers: 0-50 @ 0.58; 51-100 @ 0.68; 101-200 @ 0.83; 201-350 @ 1.25 EGP).
- **Test Inputs**:
- Reading *M_1 = 12,450.0\text{ kWh}*
- Reading *M_2 = 12,620.0\text{ kWh}*
- Consumption *\Delta C = 170.0\text{ kWh}*
- **Expected Output Breakdown**:
- Tier 1 Allocation: *50.0\text{ kWh} \times 0.58 = 29.00\text{ EGP}*
- Tier 2 Allocation: *50.0\text{ kWh} \times 0.68 = 34.00\text{ EGP}*
- Tier 3 Allocation: *70.0\text{ kWh} \times 0.83 = 58.10\text{ EGP}*
- Total Calculated Cost = *29.00 + 34.00 + 58.10 = 121.10\text{ EGP}*
- Persisted snapshot contains exact JSON breakdown matching above numbers.

15.5 Perceptual Input (Voice & OCR) Verification Tests

TC-VOICE-001: Speech Recognition Confirmation Gate

- **Test Steps**:

1. Trigger voice input button.
2. Speak: "*Five thousand four hundred point five*".

3. Observe speech parsing.

- **Expected Results**:
- Extracted number *5400.5* displayed in editable confirmation modal.
- Value is NOT written to database until user explicitly taps "Confirm".
- If user taps "Cancel", buffer is discarded with zero side-effects.

TC-OCR-001: Optical Disambiguation of Multiple Numeric Blocks

- **Test Steps**:

1. Upload test image containing both meter serial (`No. 849201`) and meter display (`01428.5 kWh`).
2. Execute OCR pipeline.

- **Expected Results**:
- System detects multiple candidates.
- System presents disambiguation modal showing image thumbnails.
- User taps `01428.5` .
- Value populated into confirmation box; saved with `reading_source = 'ocr'` .

16. Requirements Traceability Matrix (RTM)

The following matrix provides end-to-end forward and backward traceability linking Functional Requirements, Formal Use Cases, Logical Data Entities, Acceptance Test Cases, Implementation Priority (MoSCoW), and Verification Methods:

Requirement ID	Requirement Description	Related Use Case	Related Entity	Acceptance Test ID	Priority	Verification Method
FR-AUTH-001	Account Registration & Hashing	UC-01	USER	TC-AUTH-001	Must Have	Automated Test & Inspection
FR-AUTH-002	User Authentication & Tokens	UC-02	USER	TC-AUTH-002	Must Have	Test & Demonstration
FR-AUTH-003	User Sign Out & Invalidation	UC-02	USER	TC-AUTH-003	Must Have	Test
FR-AUTH-004	Password Recovery Flow	UC-03	USER	TC-AUTH-004	Should Have	Test & Demonstration
FR-AUTH-005	Account & Cascading Data Deletion	UC-15	All Entities	TC-AUTH-005	Must Have	Test & Code Inspection
FR-AUTH-006	User Data Isolation & Ownership	UC-02, UC-14	All Entities	TC-SEC-001	Must Have	Security Audit & Test
FR-METER-001	Meter Profile Management	UC-04	METER	TC-METER-003	Should Have	Demonstration
FR-METER-002	Manual Reading Ingestion	UC-04	METER_READING	TC-METER-001	Must Have	Test & Demonstration
FR-METER-003	Chronological & Lower Value Checks	UC-04	METER_READING	TC-METER-002	Must Have	Automated Unit Test
FR-METER-004	Historical Reading Modification	UC-14	METER_READING	TC-METER-004	Should Have	Test
FR-METER-005	Reading Ingestion Source Metadata	UC-04, 05, 06	METER_READING	TC-METER-005	Must Have	Inspection
FR-METER-006	Dial Rollover & Replacement Handling	UC-04	METER_READING	TC-METER-006	Could Have	Test & Analysis
FR-CALC-001	Consumption Delta (ΔkWh)	UC-07	CONSUMPTION_ESTIMATE	TC-METER-001	Must Have	Automated Unit Test
FR-CALC-002	Billing Period Association	UC-07	BILLING_PERIOD	TC-CALC-002	Must Have	Test
FR-CALC-003	Progressive Tier Cost Engine	UC-07	CONSUMPTION_ESTIMATE	TC-CALC-001	Must Have	Automated Unit Test
FR-CALC-004	Estimation Disclaimer Labeling	UC-07	UI Presentation	TC-CALC-003	Must Have	Visual Inspection
FR-CALC-005	Dynamic Recalculation on Mutation	UC-07, UC-14	CONSUMPTION_ESTIMATE	TC-CALC-004	Must Have	Test
FR-VOICE-001	Mic Hardware Permission Handling	UC-05	Hardware Interface	TC-VOICE-002	Should Have	Demonstration
FR-VOICE-002	Bilingual Egyptian Arabic/English STT	UC-05	Perception Interface	TC-VOICE-003	Should Have	Test
FR-VOICE-003	Numeric Token Parsing & Decimals	UC-05	Domain Parser	TC-VOICE-004	Should Have	Automated Unit Test

Requirement ID	Requirement Description	Related Use Case	Related Entity	Acceptance Test ID	Priority	Verification Method
FR-VOICE-004	Voice Review & Confirmation Gate	UC-05	METER_READING	TC-VOICE-001	Must Have	Test & Demonstration
FR-OCR-001	Camera & Gallery Image Picker	UC-06	Hardware Interface	TC-OCR-002	Should Have	Demonstration
FR-OCR-002	OCR Text Detection Execution	UC-06	Perception Interface	TC-OCR-003	Should Have	Test
FR-OCR-003	Candidate Filtering & Disambiguation	UC-06	Domain Parser	TC-OCR-001	Should Have	Test & Demonstration
FR-OCR-004	OCR Review Modal & Source Tag	UC-06	METER_READING	TC-OCR-004	Must Have	Demonstration
FR-CHART-001	Historical Consumption Bar Charts	UC-08	Visual Analytics	TC-CHART-001	Must Have	Demonstration
FR-CHART-002	Appliance Share Donut/Pie Chart	UC-08	Visual Analytics	TC-CHART-002	Should Have	Demonstration
FR-CHART-003	Period-over-Period Indicators	UC-08	Visual Analytics	TC-CHART-003	Should Have	Test
FR-CHART-004	Dynamic Empty-State Presentations	UC-08	UI Presentation	TC-CHART-004	Must Have	Visual Inspection
FR-BUDGET-001	Budget Creation & Configuration	UC-09	BUDGET	TC-BUDGET-001	Must Have	Test
FR-BUDGET-002	Real-Time Spending Tracking	UC-09	BUDGET	TC-BUDGET-002	Must Have	Automated Test
FR-BUDGET-003	80% and 100% Threshold Warnings	UC-09	BUDGET	TC-BUDGET-003	Must Have	Test & Demonstration
FR-BUDGET-004	Month-End Run-Rate Projection	UC-09	BUDGET	TC-BUDGET-004	Should Have	Automated Unit Test
FR-APP-001	Appliance Inventory Management	UC-10	APPLIANCE	TC-APP-002	Must Have	Test
FR-APP-002	Operating Hours & Duty Cycles	UC-10	APPLIANCE_USAGE_SCHEDULE	TC-APP-001	Must Have	Automated Unit Test
FR-APP-003	Appliance Energy Computation (E)	UC-10	APPLIANCE	TC-APP-003	Must Have	Automated Unit Test
FR-APP-004	AC Active Power ($P = VI\cos\phi$)	UC-10	APPLIANCE	TC-APP-004	Must Have	Automated Unit Test
FR-APP-005	Standby Vampire Power Modeling	UC-10	APPLIANCE	TC-APP-005	Could Have	Test
FR-REC-001	Deterministic Optimization Engine	UC-12	RECOMMENDATION	TC-REC-001	Should Have	Analysis & Test
FR-REC-002	High-Drain Appliance Detection	UC-12	RECOMMENDATION	TC-REC-002	Should Have	Automated Unit Test
FR-REC-003	What-If Scenario Simulator	UC-11	USAGE_SCENARIO	TC-REC-003	Should Have	Test & Demonstration
FR-TARIFF-001	Structured Tariff Data Model	UC-13	TARIFF_CONFIGURATION	TC-TARIFF-001	Must Have	Code Inspection
FR-TARIFF-002	Historical Version Retention	UC-13	TARIFF_CONFIGURATION	TC-TARIFF-002	Must Have	Test & Inspection

Requirement ID	Requirement Description	Related Use Case	Related Entity	Acceptance Test ID	Priority	Verification Method
FR-TARIFF-003	Controlled Update Delivery	UC-13	TARIFF_CONFIGURATION	TC-TARIFF-003	Must Have	Demonstration
FR-TARIFF-004	Non-Invention of Official Rates	UC-13	TARIFF_TIER	TC-TARIFF-004	Must Have	Audit & Inspection
FR-PRIV-001	Row-Level Tenant Partitioning	UC-02, 14, 15	All Entities	TC-SEC-001	Must Have	Security Audit
FR-PRIV-002	Local Storage Keystore Encryption	UC-02	Storage Layer	TC-SEC-002	Must Have	Security Audit
FR-PRIV-003	Ephemeral Perception Processing	UC-05, 06	Perception Layer	TC-PRIV-001	Must Have	Code Audit
FR-SET-001	Arabic/English Bilingual Mirroring	Settings	APPLICATION_SETTINGS	TC-LOC-001	Must Have	Visual Inspection
FR-SET-002	Egyptian Currency & Digit Format	Settings	APPLICATION_SETTINGS	TC-LOC-002	Should Have	Demonstration
FR-SET-003	Comprehensive Audit Log & Filter	UC-14	METER_READING	TC-METER-004	Must Have	Demonstration

Verification Methods:

- **Test**: Automated or scripted software test execution with discrete inputs and expected outputs.
- **Demonstration**: Observable operational verification of UI behavior and workflows.
- **Inspection**: Direct visual examination of code, database schema, or UI strings.
- **Analysis**: Mathematical verification of calculation formulas and algorithmic properties.

17. Future Enhancements

The following features are intentionally segregated from the baseline release and categorized as potential roadmap candidates for future product revisions:

1. **Direct Utility Clearinghouse Integration**: Secure API integration with the Egyptian Electricity Holding Company (EEHC) to import certified official monthly meter readings and billing statements.
2. **Prepaid Smart Card NFC Integration**: Direct reading of balance and accumulated kilowatt-hours from contactless Egyptian smart utility cards via mobile NFC transceivers.
3. **Automated Bill Payment Gateways**: Integration of national retail payment channels (Fawry, Meeza, Vodafone Cash, InstaPay) enabling users to settle bills directly within the application.
4. **IoT Smart Energy Monitor Coupling**: Telemetry integration with Wi-Fi/Zigbee current transformers (CT clamps) installed inside consumer distribution panels for continuous real-time power logging.
5. **Multi-User Family Account Delegation**: Role-based access control allowing household members to view consumption dashboards while restricting billing configuration to the primary account holder.
6. **Machine Learning Load Disaggregation (NILM)**: Advanced neural network pattern analysis estimating individual appliance activity from overall meter load curves without manual appliance scheduling.
7. **Administrative Web Portal for Tariff Authorities**: Cloud management portal enabling regulatory authorities to broadcast certified, cryptographically signed tariff revisions directly to mobile clients.

18. Open Questions, Assumptions, Dependencies, and Constraints

18.1 Storage Architecture Analysis & Evaluation (Trade-off Matrix)

The data storage architecture is **formally undecided** as a project constraint. Below is an exhaustive comparative evaluation of the three architectural archetypes across ten core engineering dimensions:

#	Evaluation Dimension	Option A: Local-Only (SQLite / Hive on Dev)	Option B: Cloud-Only (Remote Cloud DB API)	Option C: Hybrid Sync (Local SQLite + Cloud Sync)
1	Privacy & Data Sovereignty	Maximum. Data never leaves physical phone.	Medium. Relies on cloud provider trust.	High. Data stored locally; encrypted during transit.
2	Multi-Device Access	Impossible. Records trapped on single OS.	Seamless across web, iOS, and Android.	Supported via automated background synchronization.
3	Offline Behavior	100% full read/write completely offline.	App crippled or read-only without network.	Full offline read/write; queues sync events.
4	Backup & Recovery	Poor. Lost if phone is destroyed or stolen.	Automated. Restored on any new device.	Robust. Cloud maintains backup of local database.
5	Implementation Complexity	Low. Pure Flutter client data layer.	Medium. Requires backend & auth API.	High. Requires conflict resolution engine & queues.
6	Auth & Access Control	Simple local PIN or biometrics.	Complex JWT / OAuth2 and Row-Level Security	Multi-tiered auth; local cache tied to cloud token.
7	Sync Conflicts	None. Single writer.	None. Single source of truth on server.	Complex. Requires CRDT or "last-write-wins" logic.
8	Data Consistency	Immediate / ACID local consistency.	Immediate / ACID on server.	Eventual consistency across devices.
9	Operating Cost	$0 recurring server hosting expense.	Monthly cloud DB, API gateway & egress fees.	Ongoing cloud backend hosting and egress fees.
10	Maintainability	High. No backend infrastructure to fix.	Medium. Requires API versioning & DevOps.	Lower. Must maintain client sync and backend services.

Architectural Recommendation

- **Recommended Strategy**: **Option C: Hybrid Storage Architecture (Offline-First with Remote Cloud Synchronization)**.
- **Architectural Rationale**: A personal utility application in Egypt must guarantee uninterrupted offline availability due to intermittent mobile data connections in residential utility basements. However, because the product mandates **individual accounts, private user profiles, and future multi-device access**, a purely local model is inadequate for long-term data backup. The hybrid model utilizes a local SQLite database (via Flutter `sqflite` or `drift`) as the primary cache, abstracting data operations through the Repository Pattern. An asynchronous background synchronization service pushes encrypted mutations to a cloud database (e.g., Supabase PostgreSQL with Row-Level Security or Firebase Cloud Firestore) whenever network connectivity is verified.
- **Approval Status**: ⚠️ **PENDING STAKEHOLDER SIGN-OFF**. The final selection between Option A, B, and C remains an open decision requiring explicit sponsor confirmation.

18.2 Open Architectural Questions

Question ID	Architectural Topic	Technical Description & Impact	Blocking Status
OQ-ARCH-01	Final Storage Selection	Final sign-off required on Local-Only vs Cloud-Only vs Hybrid Sync architecture.	Blocks Backend Setup; Does Not Block Local UI / Calculations.

Question ID	Architectural Topic	Technical Description & Impact	Blocking Status
OQ-ARCH-02	Cloud Provider Selection	If Cloud/Hybrid chosen, select platform: Supabase (PostgreSQL RLS), Firebase, or Custom Node.js/PostgreSQL REST API.	Deferred to Sprint 2 Architecture Review.
OQ-ARCH-03	OCR Engine Location	Final determination between on-device Google ML Kit (free, offline, private) vs Cloud Vision API (higher accuracy on curved glass dials, network required).	Does not block UI; abstracted behind `IOCREngine`.
OQ-ARCH-04	Speech Recognition Engine	Select between Native OS Speech Recognizer (`speech_to_text`) vs Cloud Whisper API for Egyptian Arabic colloquial dialect accuracy.	Does not block UI; abstracted behind `ISpeechEngine`.
OQ-ARCH-05	Remote Tariff Delivery	Protocol for updating official tariffs: Static JSON served via GitHub/S3 CDN vs Firebase Remote Config vs Dedicated Admin API.	Deferred to Tariff Subsystem Milestone.

18.3 Open Business & Regulatory Questions

Question ID	Domain Topic	Description & Impact	Blocking Status
OQ-BIZ-01	Account Verification Policy	Determine whether email verification (via OTP code or magic link) is mandatory before granting access, or if immediate access is permitted.	Blocks Authentication Gate Finalization.
OQ-BIZ-02	Official Fixed Customer Fees	Egypt utility bills frequently incorporate nominal fixed monthly customer service charges (خدمة العملاء), stamps, and municipal taxes. Determine if these must be factored into estimations.	Non-blocking; can be modeled as additive tariff fields.
OQ-BIZ-03	Slab-Reset (Surcharge) Modeling	Certain Egyptian commercial/residential tariffs implement a "slab-reset" rule (where exceeding *650\text{ kWh}* resets earlier low tiers to higher flat rates). Confirm if this rule must be active in Version 1.0.0.	Requires formal confirmation from EgyptERA tariff schedules.
OQ-BIZ-04	Minimum Supported OS Versions	Confirm minimum target devices (Android 10+ / iOS 14+ proposed; evaluate whether legacy Android 8.0/9.0 devices must be accommodated).	Non-blocking baseline established.

18.4 Documented Project Assumptions

1. **User Trust & Agency**: Users understand that calculated monetary figures represent estimations designed for domestic planning, rather than legally binding utility invoices.
2. **Device Hardware Integrity**: Mobile devices have operational internal clocks and functioning camera/microphone hardware.
3. **Meter Accessibility**: Users have regular physical access to inspect their electricity meters.
4. **Tariff Monotonicity**: Tariff tiers are strictly non-overlapping and ordered sequentially by consumption.

18.5 Project Constraints

1. **Framework Constraint**: Development is strictly constrained to the Flutter framework to enable simultaneous Android and iOS delivery from a single repository.
2. **No Code Implementation in SRS**: This document represents a software requirements specification and does not constitute production code.
3. **Language Neutrality in Logic**: All domain business rules and calculation engines must execute identically regardless of whether the active interface language is Arabic or English.

19. Glossary of Terms

- **Active Power (*P*)**: The actual electrical power consumed by a circuit to perform useful work (measured in Watts or Kilowatts).
- **Apparent Power (*S*)**: The mathematical product of root-mean-square (RMS) voltage and current in an AC circuit (measured in Volt-Amperes, *text{VA}*).
- **Billing Period**: The specific time interval (typically calendar monthly) bounded by two dates, over which electricity consumption is measured and billed.
- **Consumption Delta (*Delta C*)**: The net quantity of electrical energy consumed between two readings, calculated as *M_{\text{current}} - M_{\text{previous}}*.
- **EEHC**: Egyptian Electricity Holding Company (الشركة القابضة لكهرباء مصر).
- **EgyptERA**: Egypt Electric Utility and Consumer Protection Regulatory Agency (جهاز تنظيم مرفق الكهرباء وحماية المستهلك).
- **EGP (ج.م)**: Egyptian Pound (الجنيه المصري), the official sovereign currency of Egypt.
- **Flutter**: Google's open-source UI software development kit used to build natively compiled cross-platform mobile applications.
- **Kilowatt-hour (*text{kWh}*)**: A standard unit of energy equivalent to one kilowatt (*1000\text{ W}*) of power sustained over a duration of one hour.
- **NILM**: Non-Intrusive Load Monitoring; physical or algorithmic methods for disaggregating total household power into individual appliance signatures.
- **OCR**: Optical Character Recognition; algorithmic conversion of captured photographic images of physical text into digital character strings.
- **Power Factor (*cos\phi*)**: In alternating current (*text{AC}*) electricity, the dimensionless ratio of real active power absorbed by the load to the apparent power flowing in the circuit (*0.0 \le \cos\phi \le 1.0*).
- **Progressive Tariff**: An electricity pricing structure where consumption is subdivided into sequential consumption brackets, with higher consumption brackets charged at progressively higher unit rates.
- **Reactive Power (*Q*)**: The power oscillating between source and reactive load components (capacitors and inductors) without performing net work (measured in Volt-Amperes Reactive, *text{VAR}*).
- **RLS**: Row-Level Security; a database security mechanism ensuring that users can only query and mutate rows where their authenticated user ID matches the row ownership column.
- **RTM**: Requirements Traceability Matrix; a systems engineering artifact establishing bidirectional mapping between functional requirements, use cases, and verification test cases.
- **Slab-Reset Tariff**: An alternative utility billing model wherein exceeding a consumption ceiling retroactively reprices all prior kilowatt-hours at a higher unified rate.
- **STT**: Speech-to-Text; software subsystem that transcribes acoustic human speech into textual string sequences.
- **Vampire / Standby Power**: Electrical energy consumed by electronic devices while switched off or in standby/idle mode.
- **WCAG 2.1 AA**: Web Content Accessibility Guidelines Level AA standards for mobile user accessibility.

End of Software Requirements Specification — Electricity Consumption Calculator (Version 1.0.0)